

Sequence data visualization and IGV

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BFX-workshop week 05

Adapted from:

Malachi Griffith, Obi Griffith, Fouad Yousif

High-Throughput Biology: From Sequence to Networks

https://github.com/griffithlab/rnaseq_tutorial_wiki/blob/master/LectureFiles/cbw-cshl/2017/IGV_Tutorial_Brief.pptx



Visualization Tools in Genomics

- there are **over 40 different genome browsers**, which to use?
- depends on
 - task at hand
 - kind and size of data
 - data privacy

HT-seq Genome Browsers



Integrative
Genome
Viewer



UCSC
Genome Browser
Cancer Genome Browser



Trackster
(part of Galaxy)

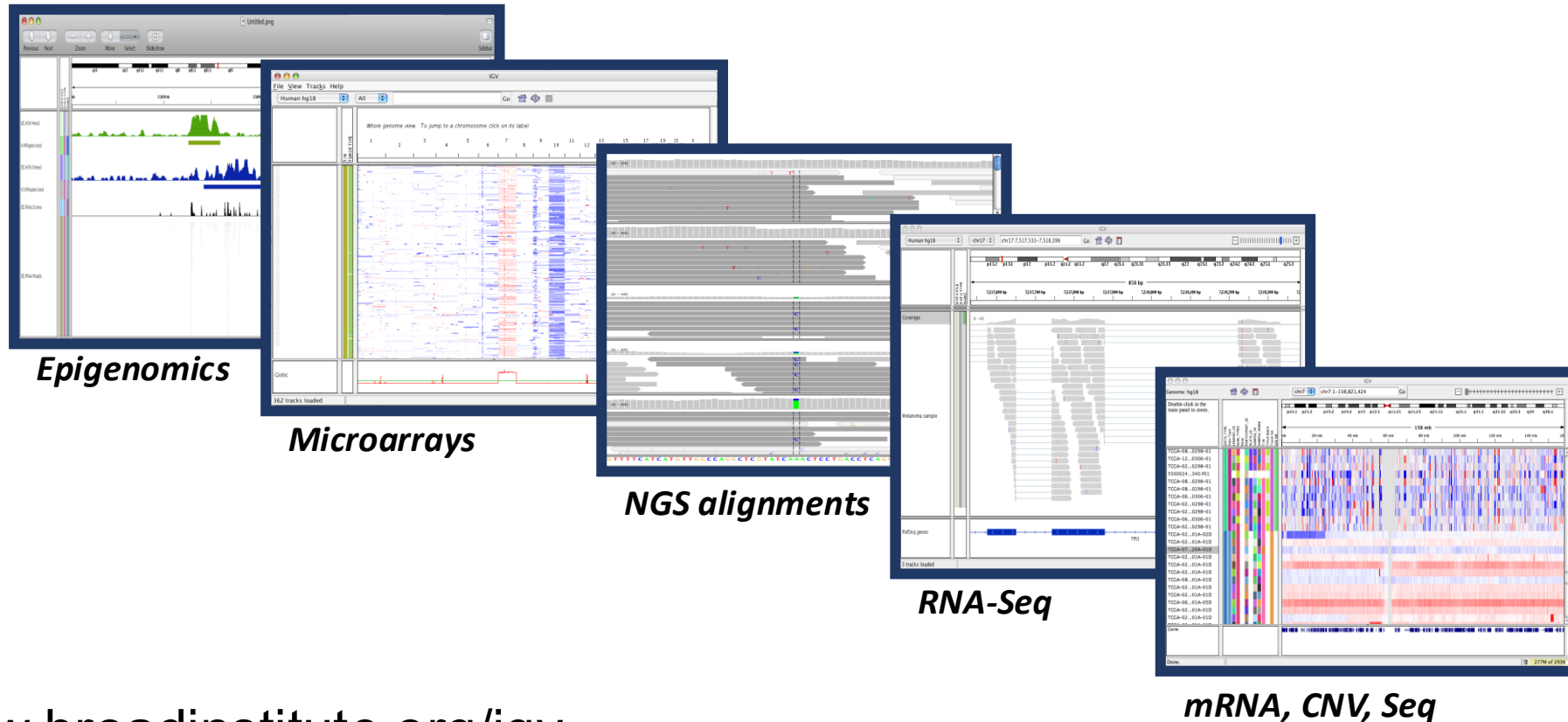


Savant
Genome
Browser

- task at hand : visualizing HT-seq reads, especially good for inspecting variants
- kind and size of data : large BAM files, stored locally or remotely
- data privacy : run on the desktop, can keep all data private
- UCSC Genome Browser has been retro-fitted to display BAM files
- Trackster is a genome browser that can perform visual analytics on small windows of the genome, deploy full analysis with Galaxy

Integrative Genomics Viewer (IGV)

Desktop application for the interactive visual exploration of integrated genomic datasets



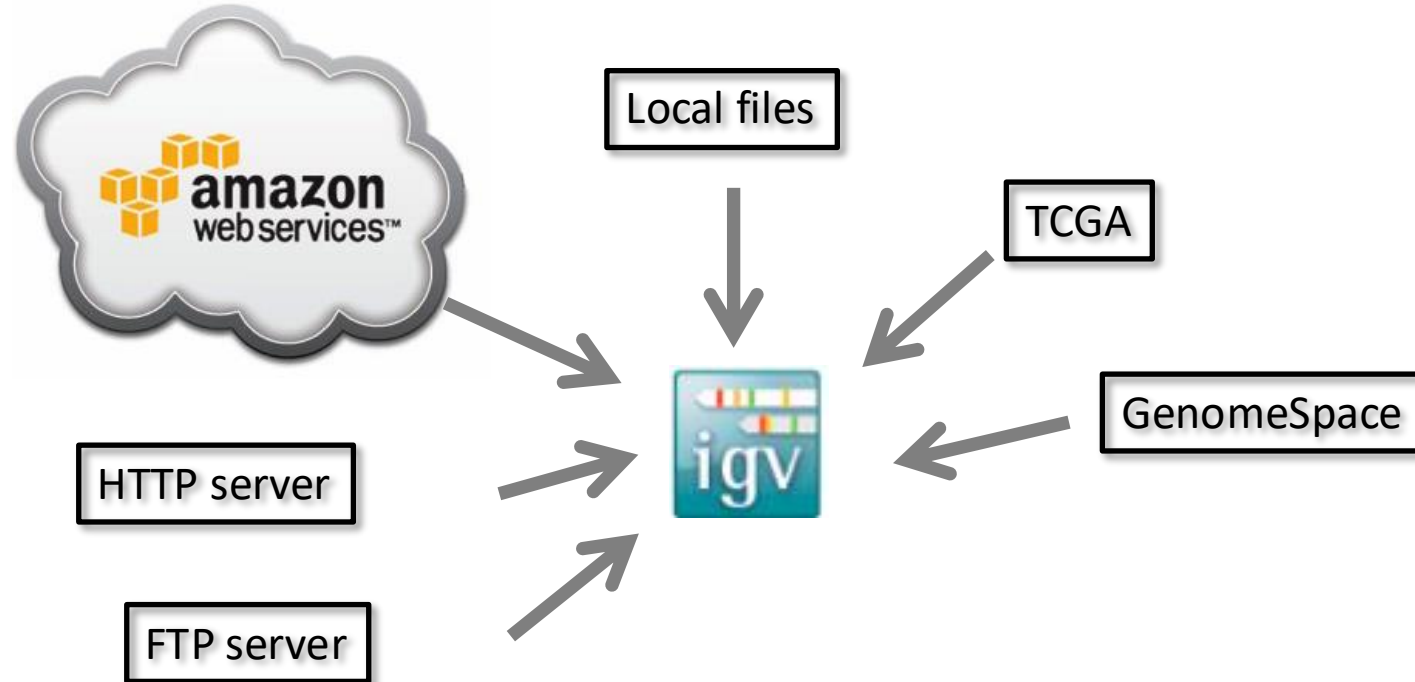
<http://www.broadinstitute.org/igv>

Features of IGV

With IGV you can...

- intuitive, easy-to-use interface
- Scales well to large data
- Integrate multiple data types
- View data from multiple locations:
 - local, remote, and “cloud-based”.
- Some automation of tasks using command-line interface

IGV data sources



- View **local** files without uploading.
- View **remote** files without downloading the whole dataset.

Using IGV: the basics

- Launch IGV
- Select a reference genome
- Load data
- Navigate through the data
 - WGS data
 - SNVs
 - structural variations

Launch IGV

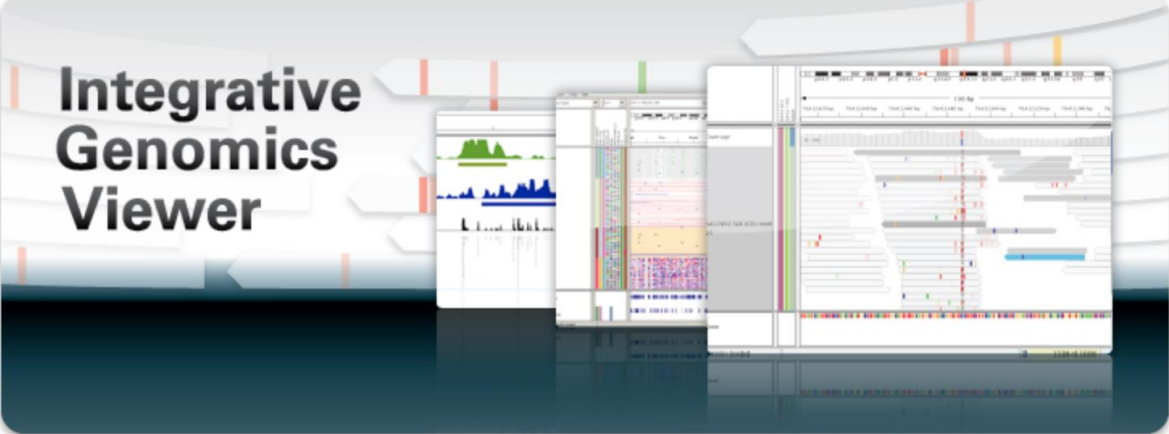
**Integrative Genomics Viewer**
v2.3.6

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Home



Integrative Genomics Viewer

Overview



The **Integrative Genomics Viewer (IGV)** is a high-performance, easy-to-use, interactive tool for the visual exploration of genomic data. It supports flexible integration of all the common types of genomic data and metadata, investigator-generated or publicly available, loaded from local or cloud sources.

IGV is available in multiple forms, including:

- the original **IGV** - a Java desktop application,
- **IGV-Web** - a web application,
- **igv.js** - a JavaScript component that can be embedded in web pages *(for developers)*

Citing IGV

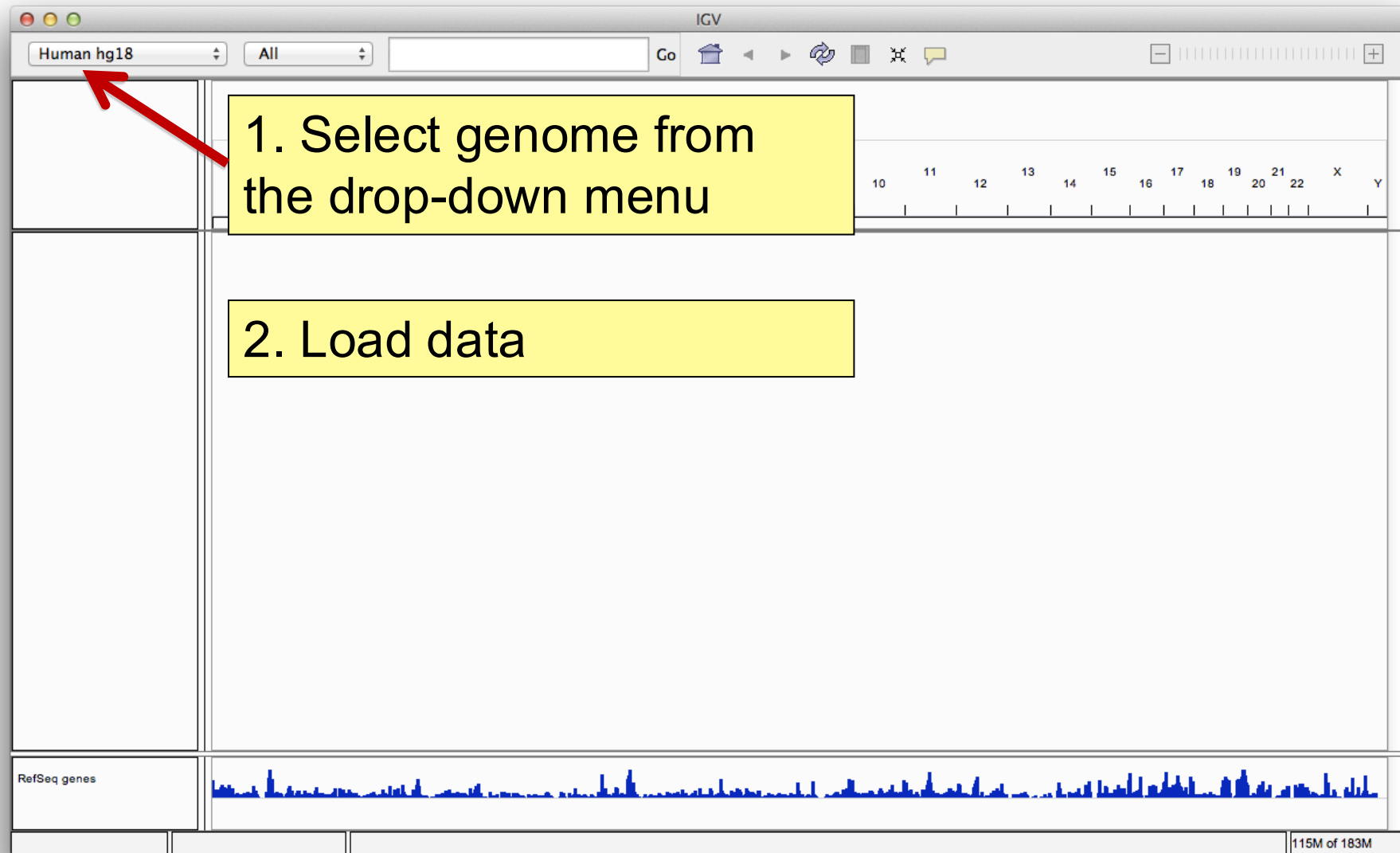
To cite your use of IGV in your publication, please reference one or more of:

James T. Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S. Lander, Gad Getz, Jill P. Mesirov. [Integrative Genomics Viewer. Nature Biotechnology 29, 24–26 \(2011\).](#) (Free PMC article [here](#)).

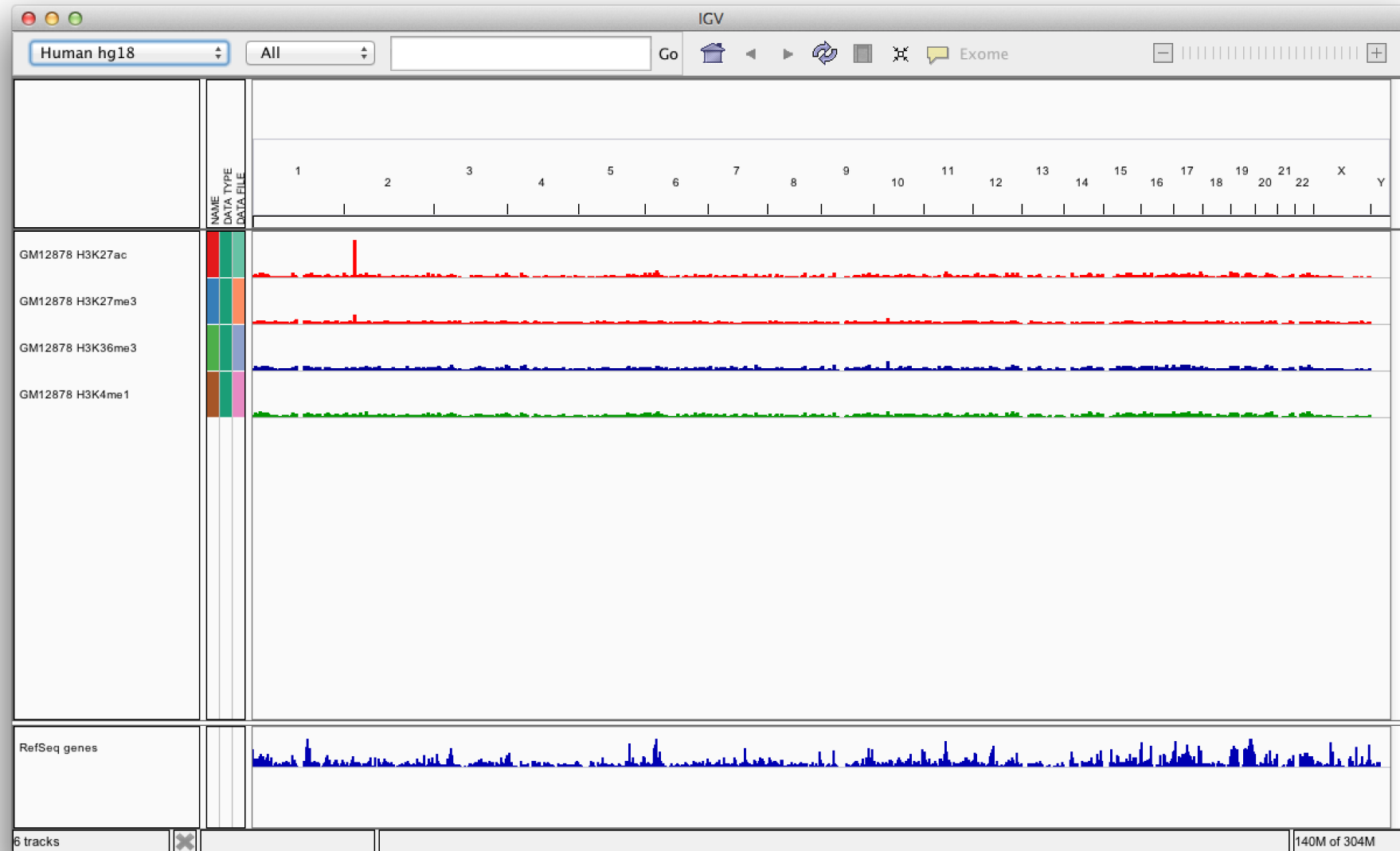
Helga Thorvaldsdóttir, James T. Robinson, Jill P. Mesirov. [Integrative Genomics Viewer \(IGV\): high-performance genomics data visualization and exploration. Briefings in Bioinformatics 14, 178-192 \(2013\).](#)



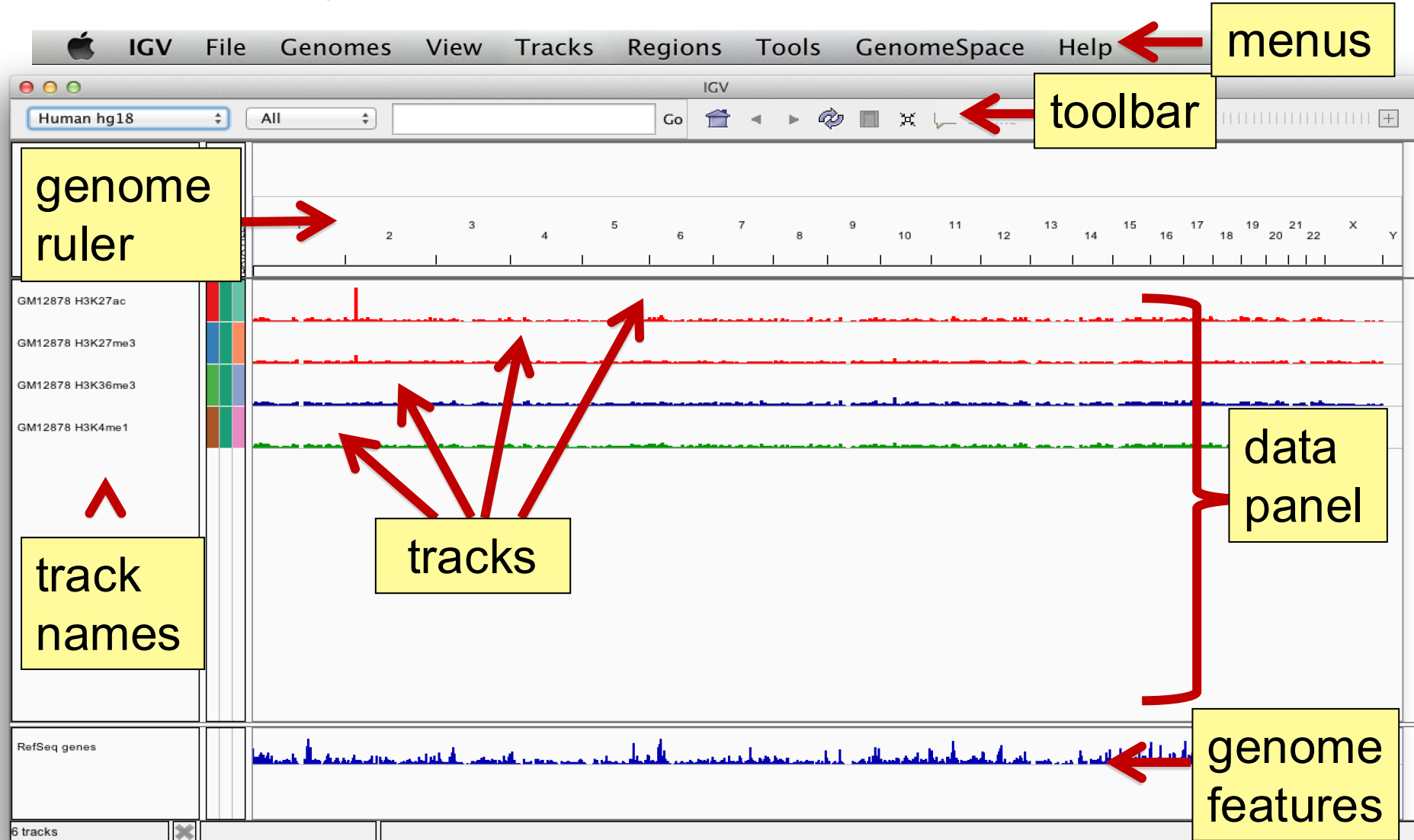
Launch IGV



Screen layout



Screen layout



File formats and track types

- The **file format** defines the track type.
- The **track type** determines the display options

- | | |
|---------------------------------------|---|
| ▪ BAM | ▪ IGV |
| ▪ BED | ▪ LOH |
| ▪ BedGraph | ▪ MAF |
| ▪ bigBed | ▪ Merged BAM File (.bam.list) |
| ▪ bigWig | ▪ MUT |
| ▪ Birdsuite Files | ▪ PSL |
| ▪ CBS | ▪ RES |
| ▪ CN | ▪ SAM |
| ▪ Cufflinks Files | ▪ Sample Information |
| ▪ Custom File Formats | ▪ SEG |
| ▪ Cytoband | ▪ SNP |
| ▪ FASTA | ▪ TAB |
| ▪ GCT | ▪ TDF |
| ▪ genePred | ▪ Track Line |
| ▪ GFF | ▪ Type Line |
| ▪ GISTIC | ▪ VCF |
| ▪ Goby | ▪ WIG |
| ▪ GWAS | |

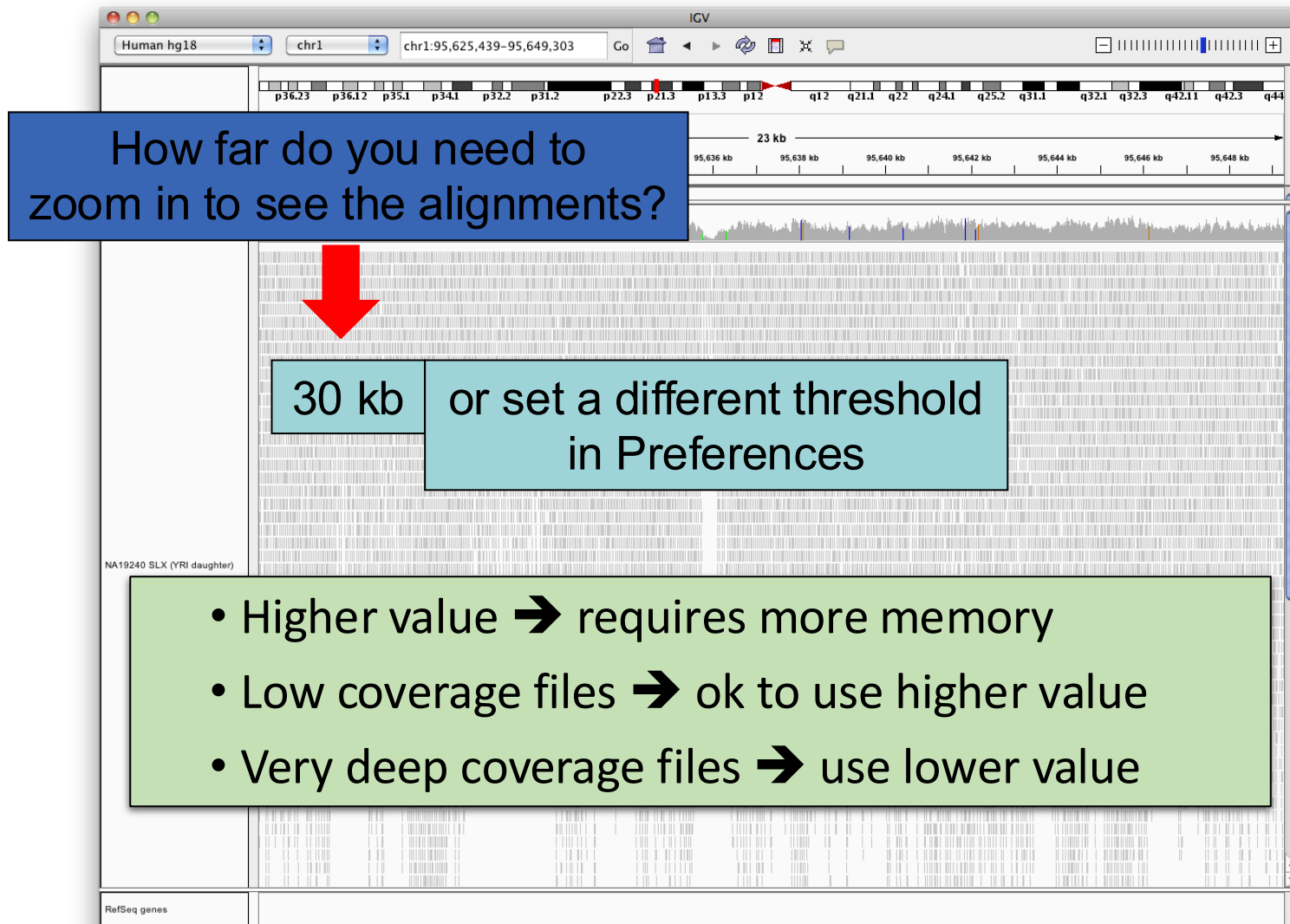
- For current list see: www.broadinstitute.org/igv/FileFormats

Viewing alignments

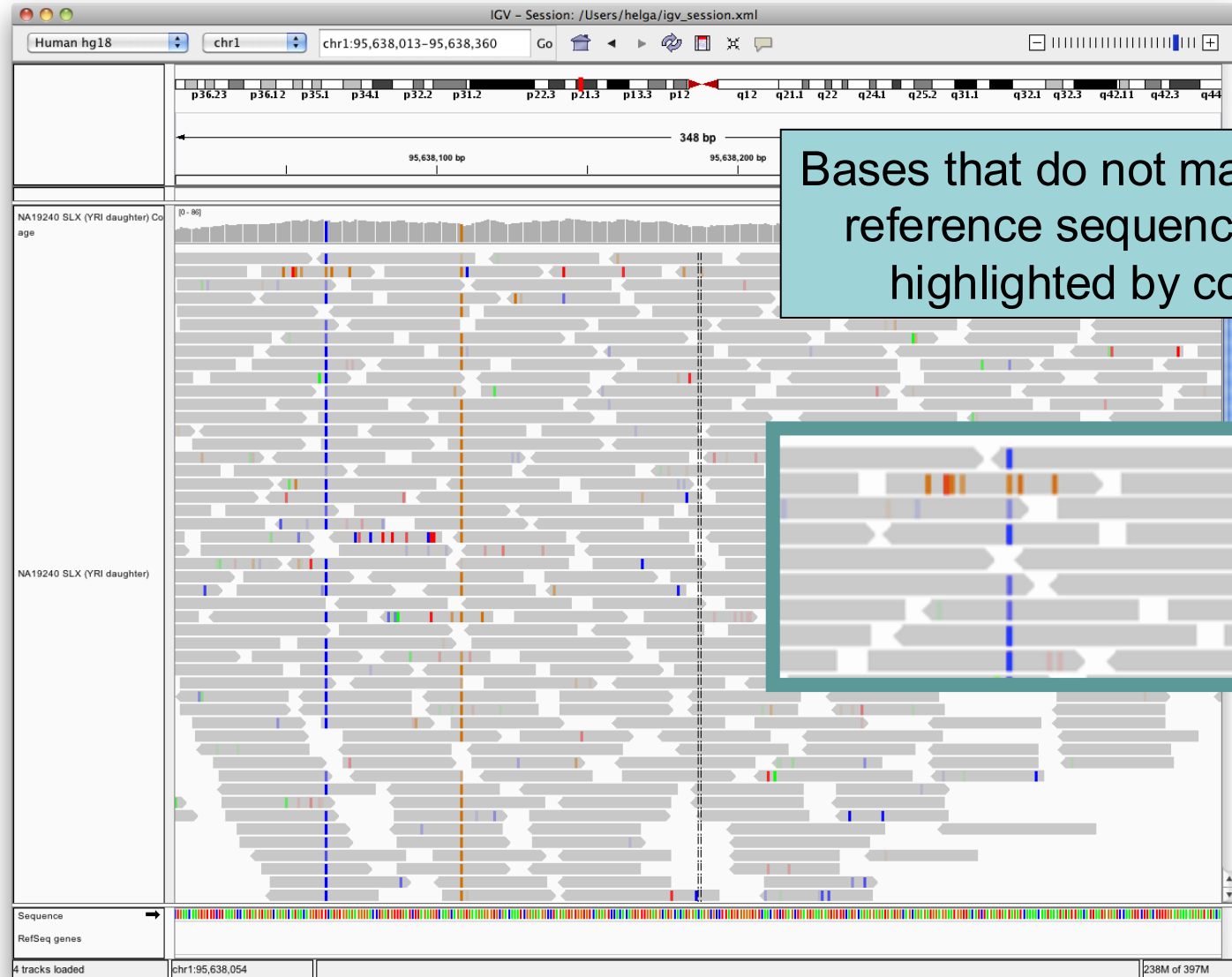
Whole chromosome view



Viewing alignments – Zoom in



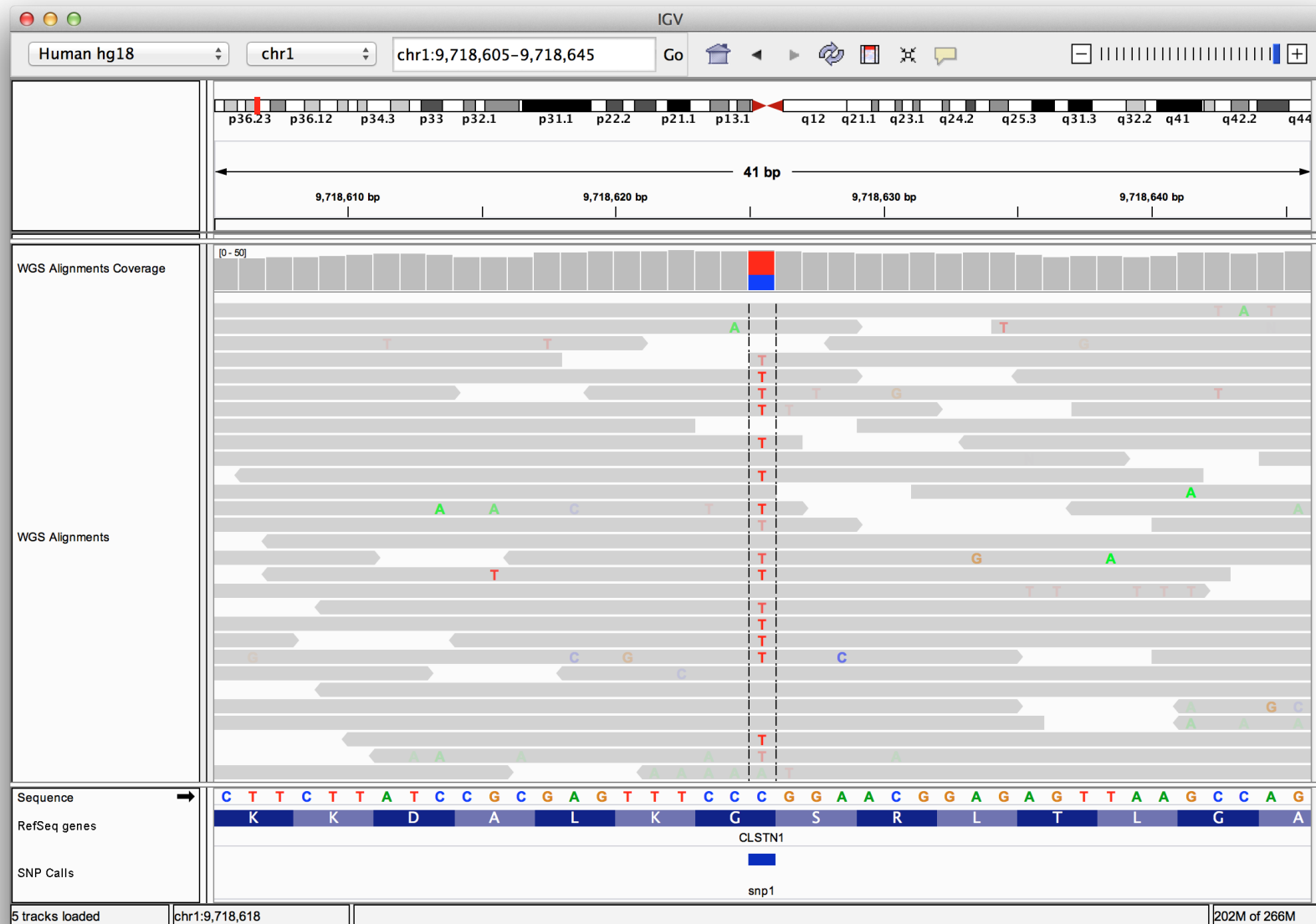
Viewing alignments – Zoom in



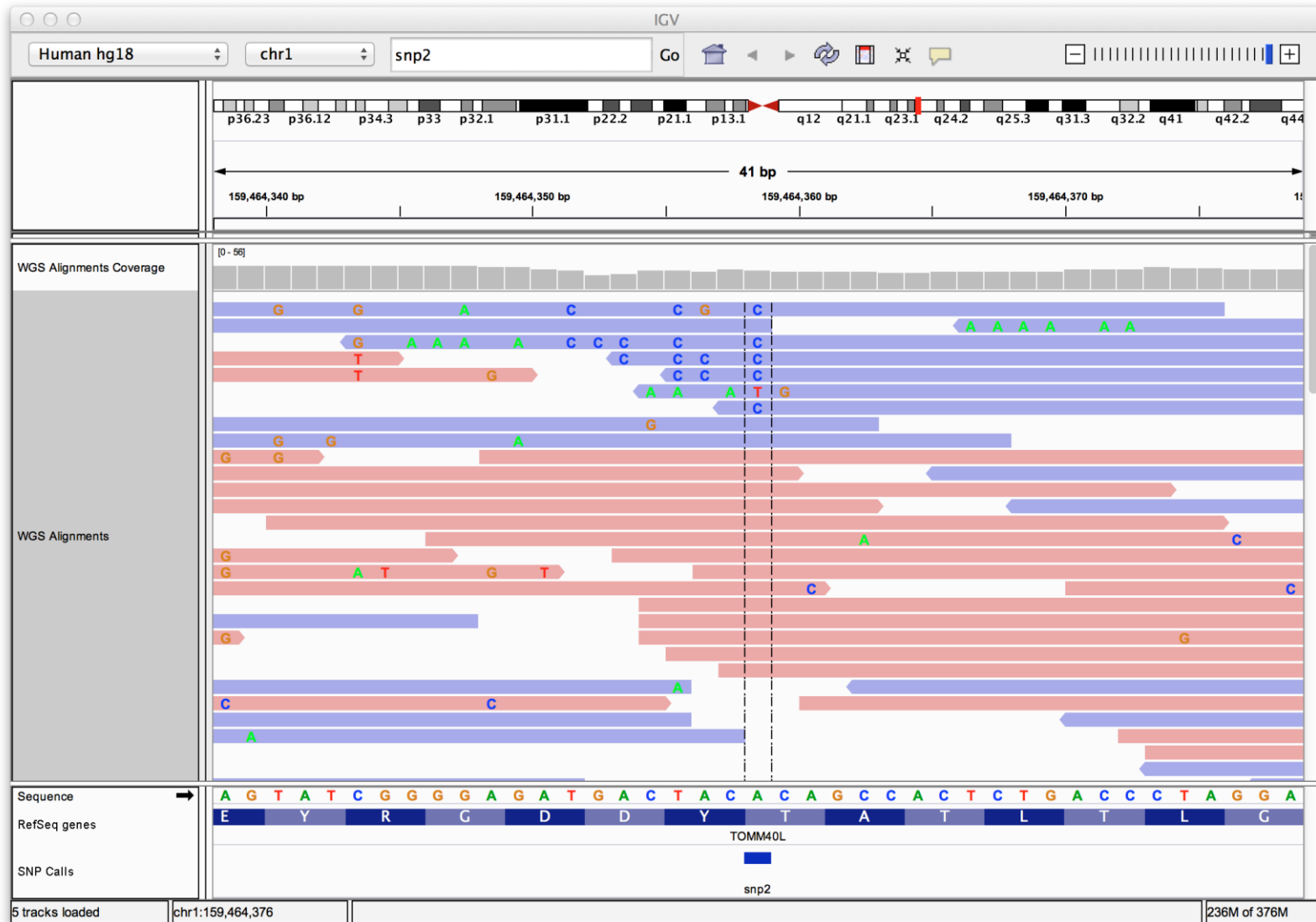
SNVs and Structural variations

- Important metrics for evaluating the validity of SNVs:
 - Coverage
 - Amount of support
 - Strand bias / PCR artifacts
 - Mapping qualities
 - Base qualities
- Important metrics for evaluating SVs:
 - Coverage
 - Insert size
 - Read pair orientation

Viewing SNPs and SNVs

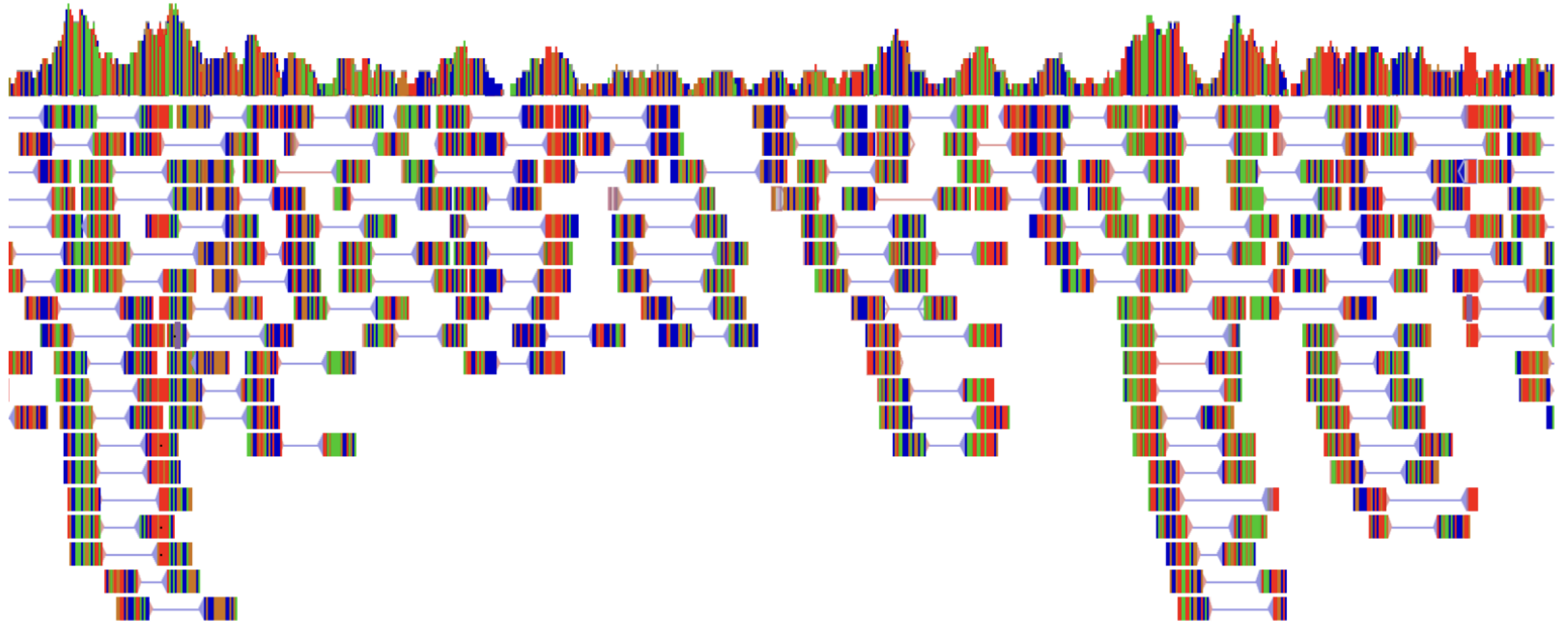


Viewing SNPs and SNVs



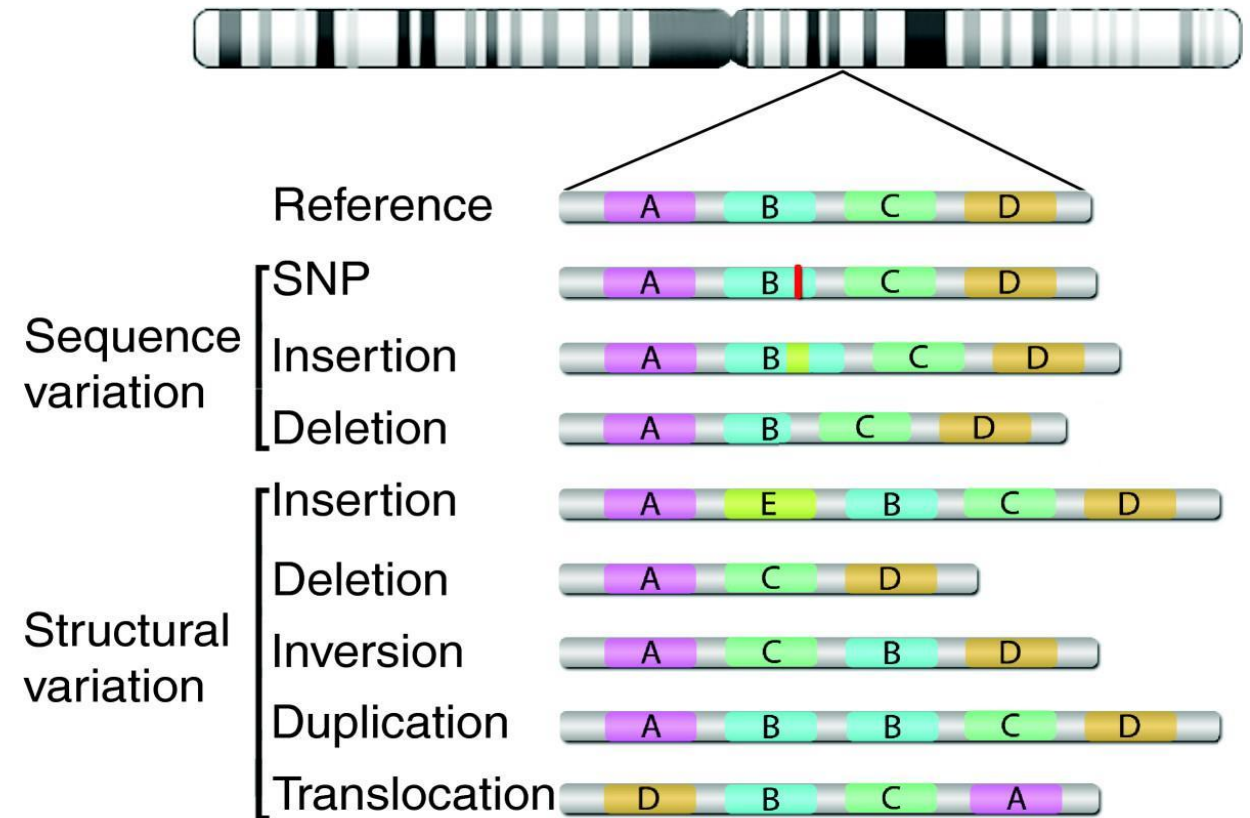
Viewing SNPs and SNVs

Viewing SNPs and SNVs



Viewing Structural Events

- Paired reads can yield evidence for genomic “structural events”, such as insertions, deletions, translocations, and inversions.
- Alignment coloring options help highlight these events based on:
 - Inferred insert size (template length)
 - Pair orientation (relative strand of pair)



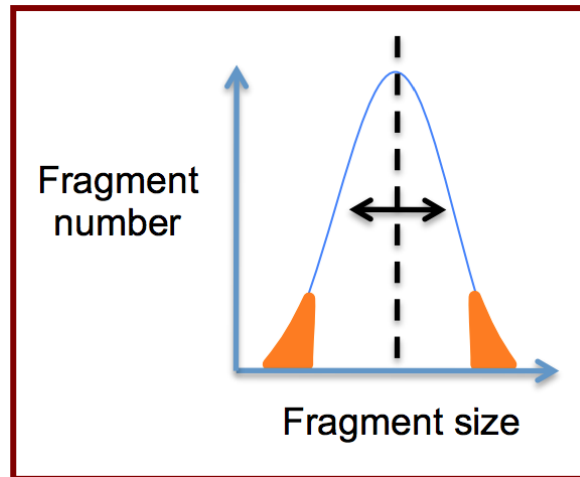
Rahim, et al. doi:10.1186/gb-2008-9-4-215

Paired-end sequencing

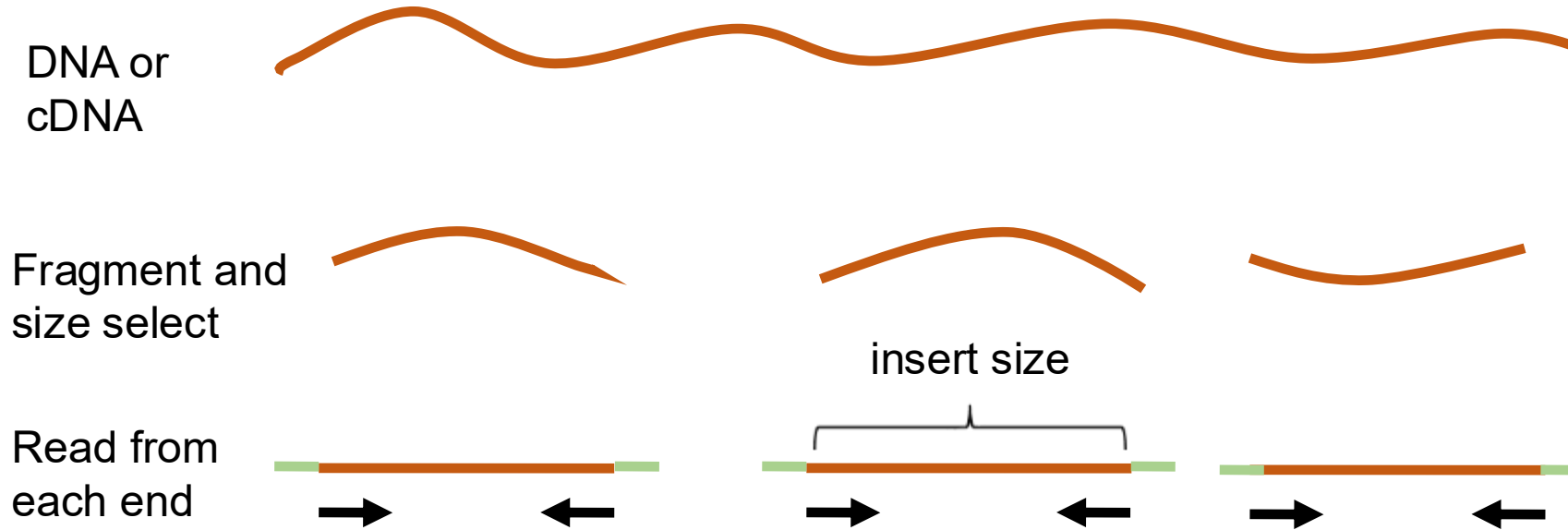
DNA or
cDNA



Fragment and
size select



Paired-end sequencing



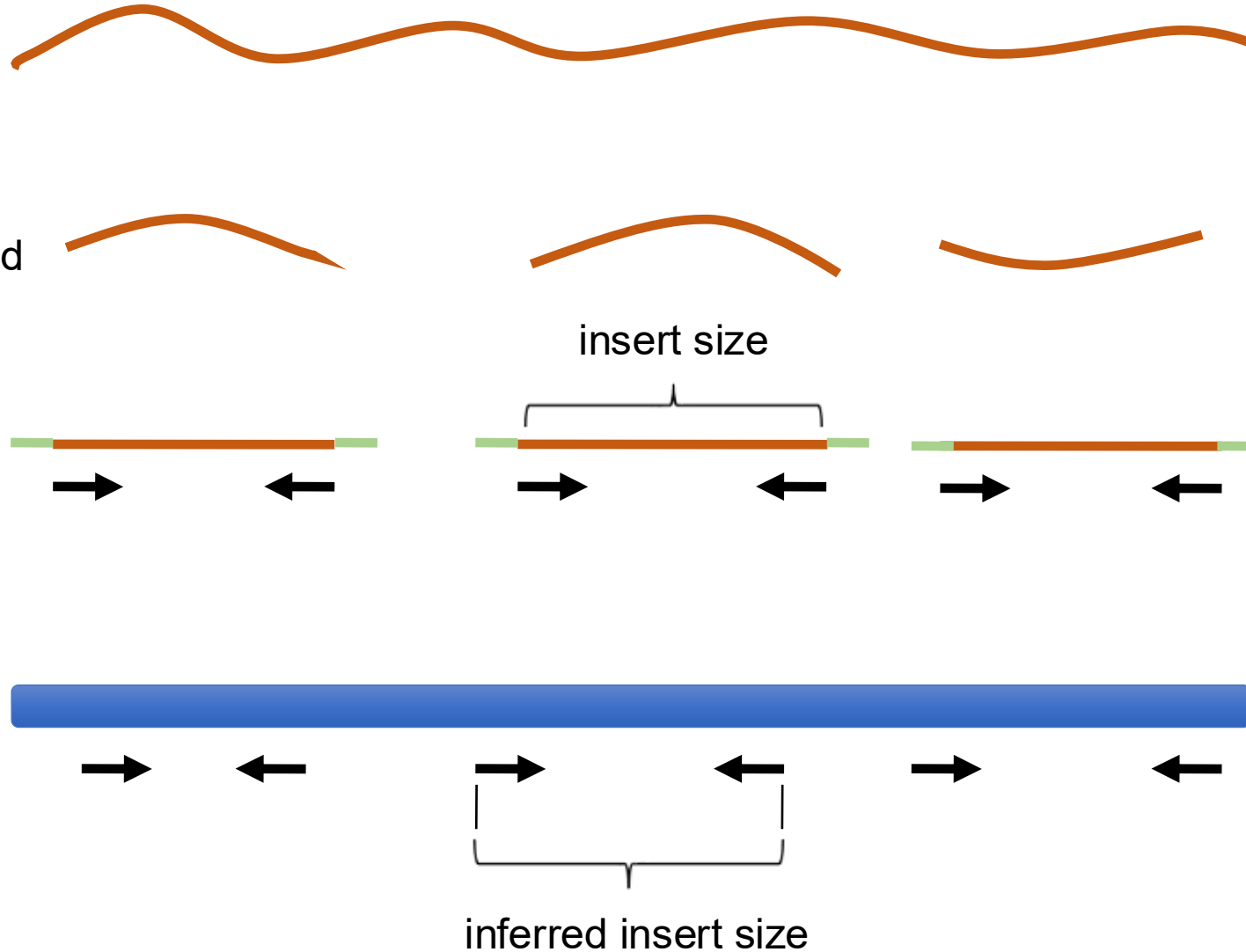
Paired-end sequencing

DNA or
cDNA

Fragment and
size select

Read from
each end

Align to
Reference



Interpreting inferred insert size

The “inferred insert size” can be used to detect structural variants including

- Deletions
- Insertions
- Inter-chromosomal rearrangements:
(Undefined insert size)

Deletion

What is the effect of a deletion on inferred insert size?

Deletion

Reference
Genome



Subject



Deletion

Reference
Genome



Subject

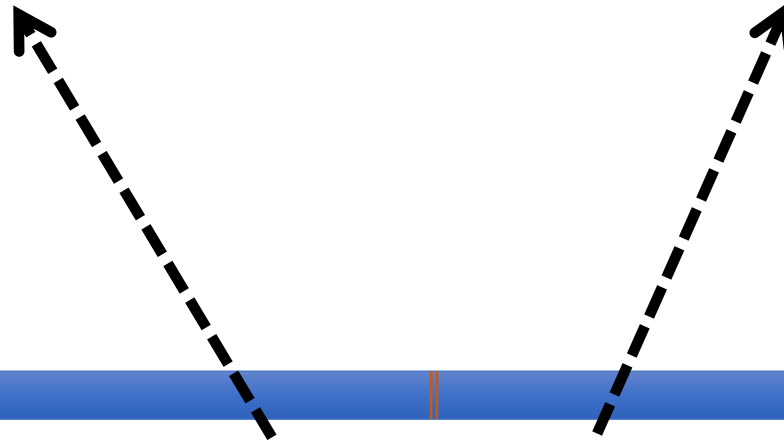
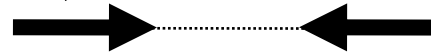


Deletion

Reference
Genome



Subject



Deletion

Inferred insert size is $>$ expected value

Reference
Genome



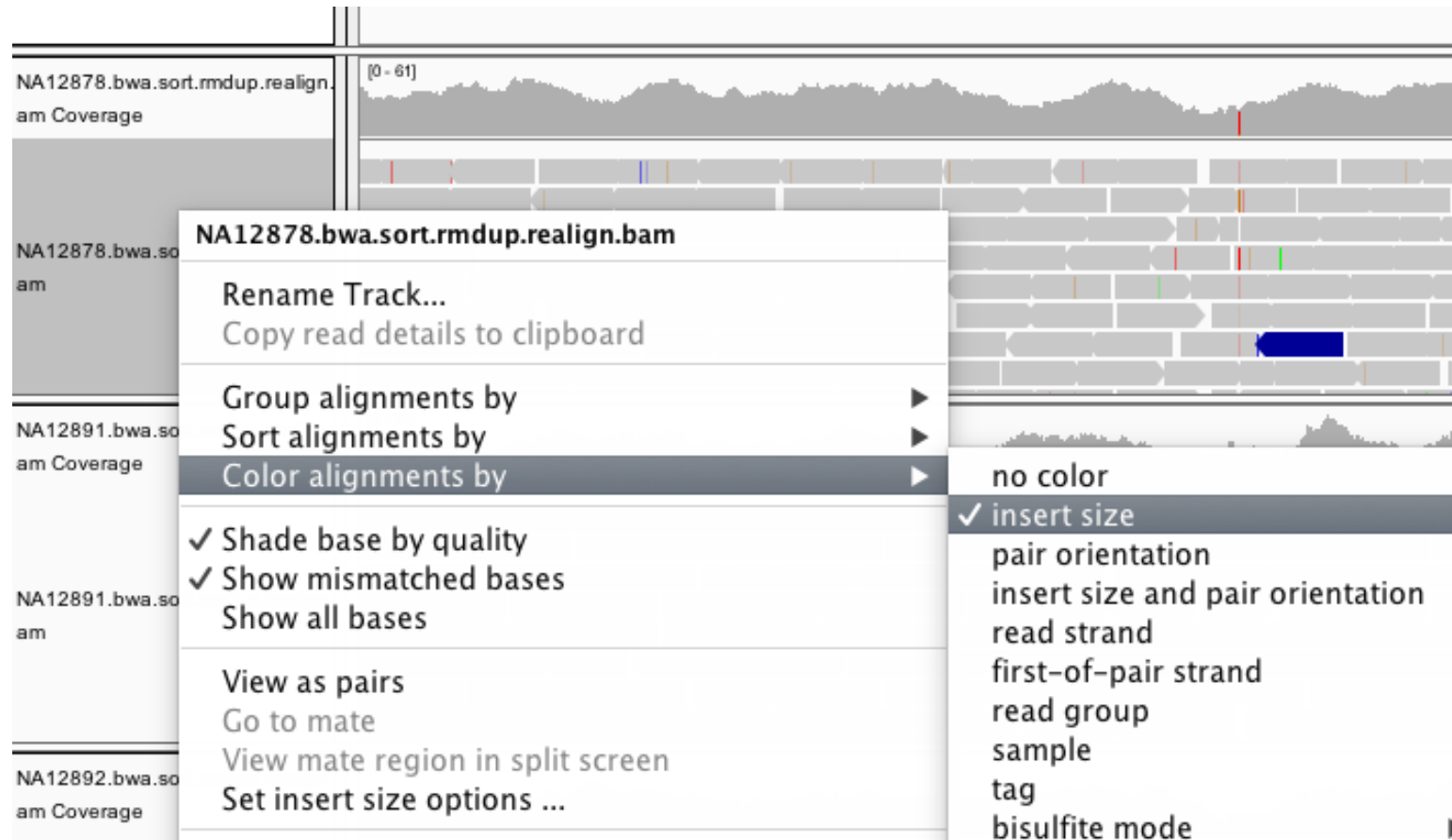
inferred insert size

Subject

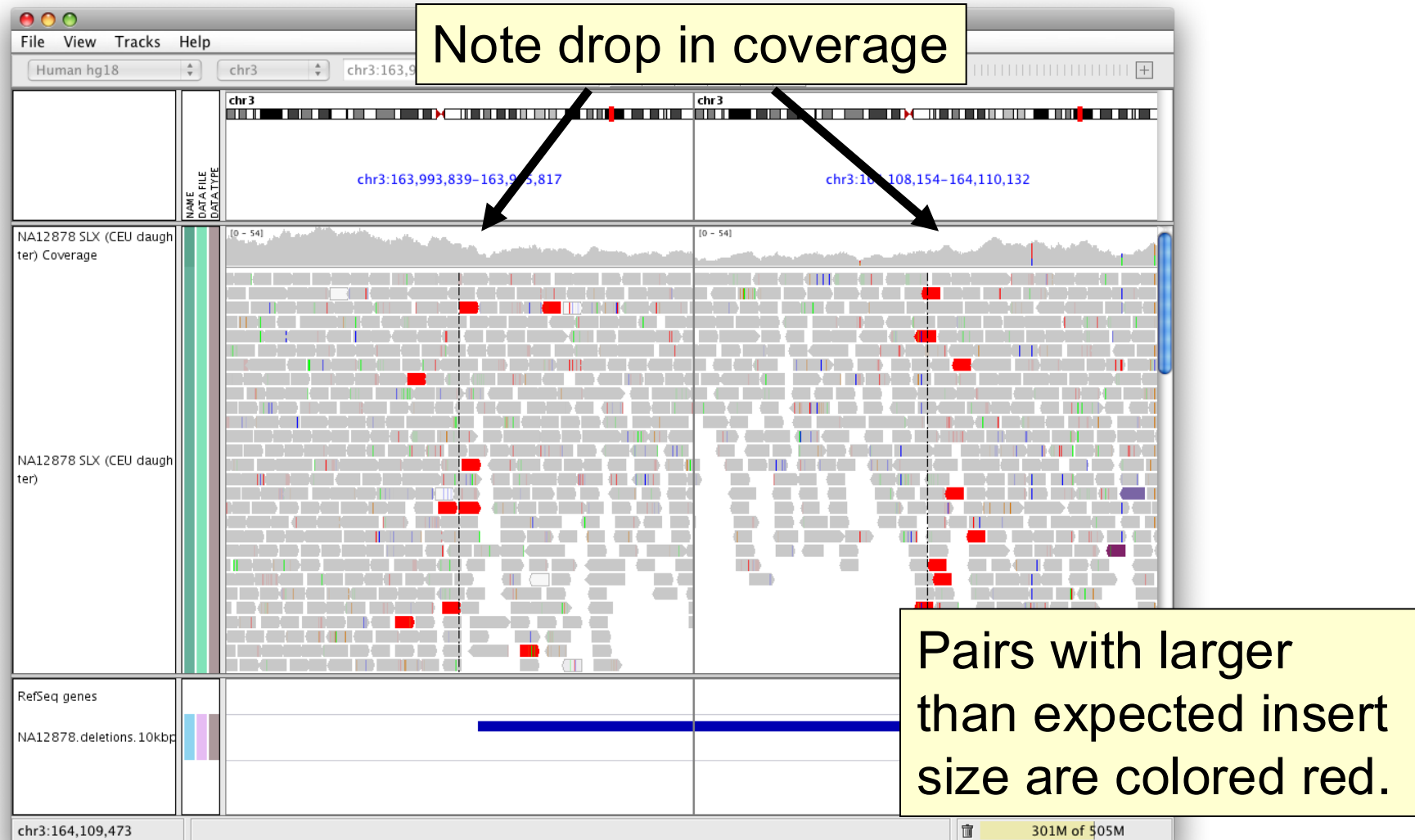


expected insert size

Color by insert size



Deletion

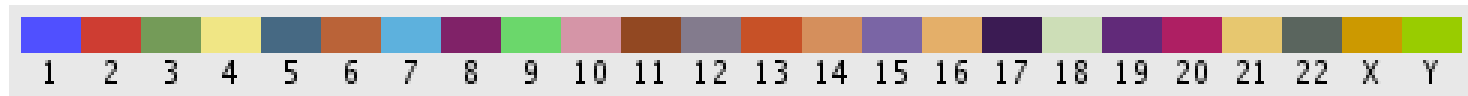


Insert size color scheme

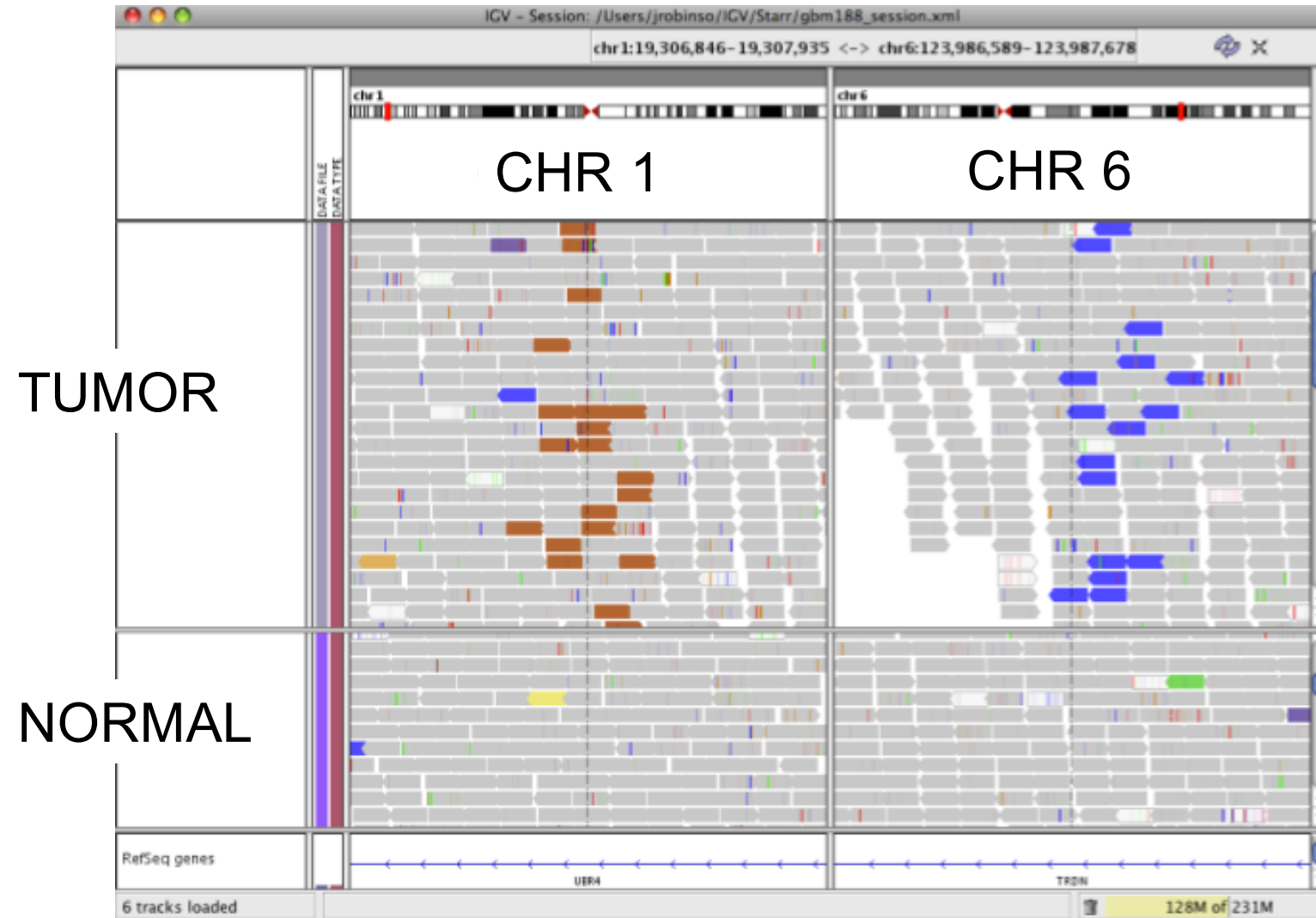
- Smaller than expected insert size:
- Larger than expected insert size:
- Pairs on different chromosomes



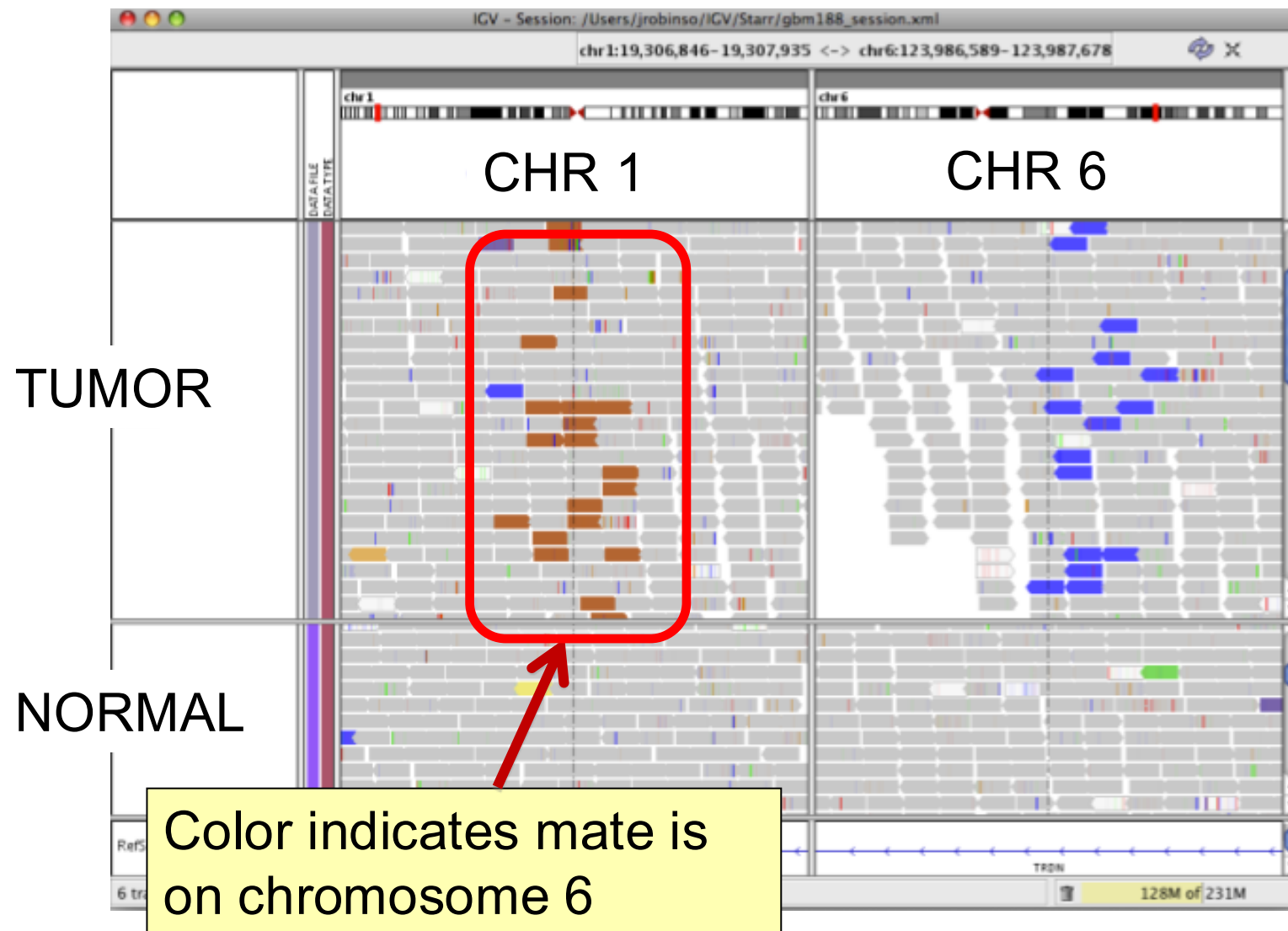
Each end colored by chromosome of its mate



Rearrangement



Rearrangement



Interpreting Read-Pair Orientations

Orientation of paired reads can reveal structural events:

- Inversions
- Duplications
- Translocations
- Complex rearrangements

Orientation is defined in terms of

- read strand, left *vs* right, *and*
- read order, first *vs* second

Inversion

Reference
genome

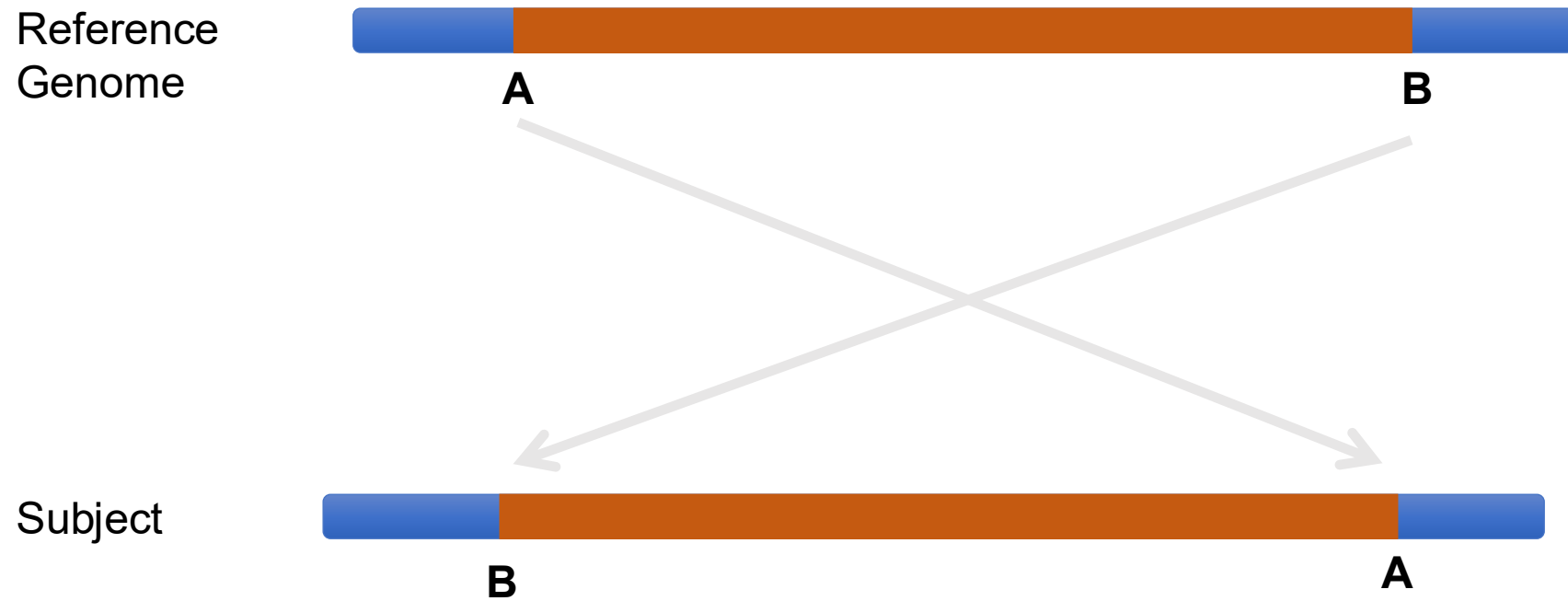


Inversion

Reference
genome



Inversion



Inversion

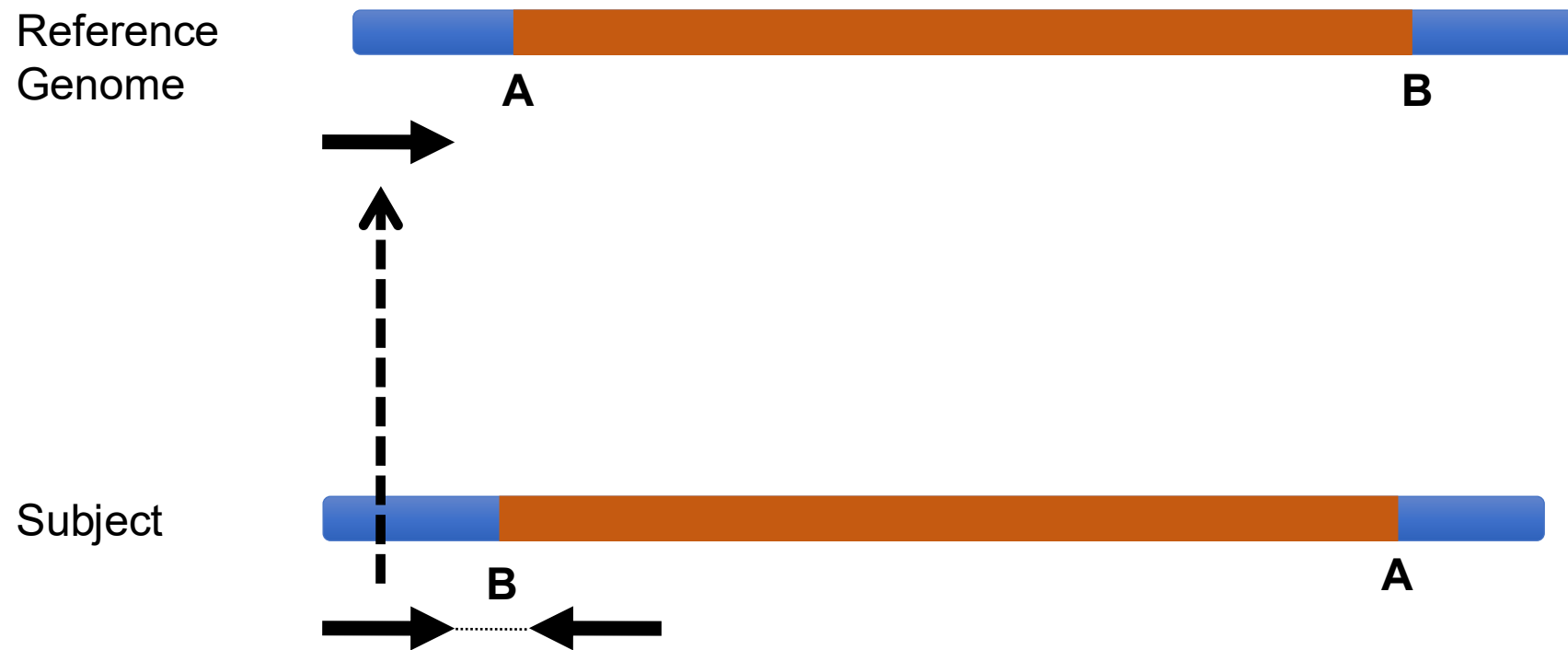
Reference
Genome



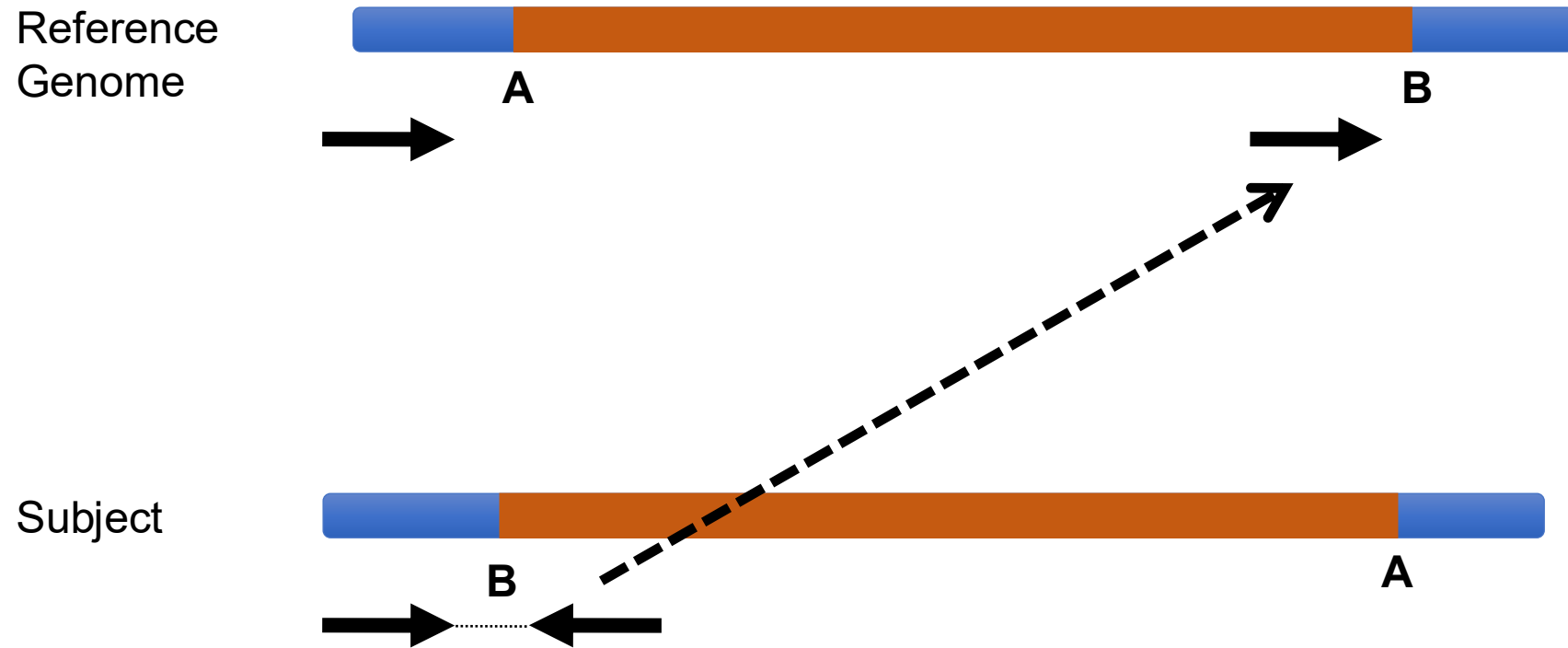
Subject



Inversion



Inversion



Inversion

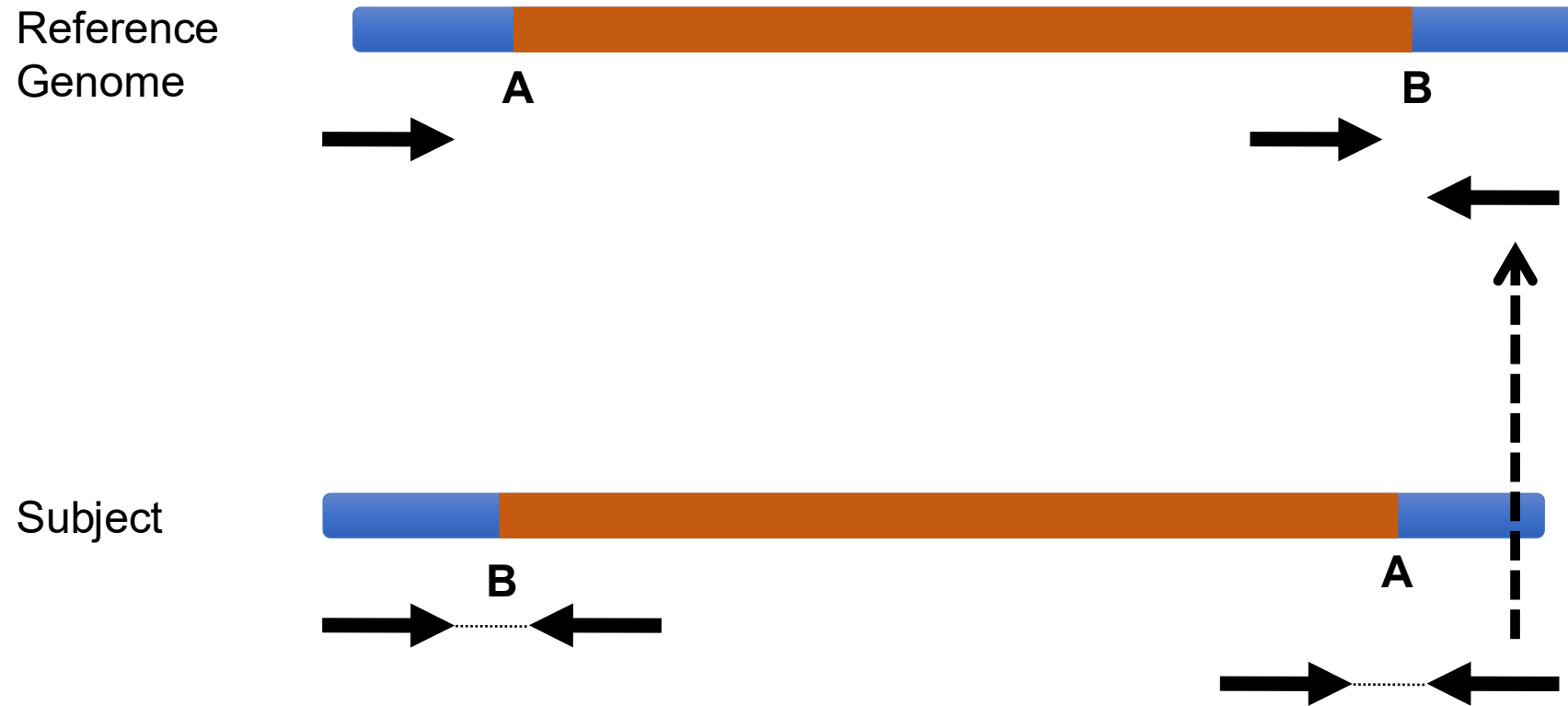
Reference
Genome



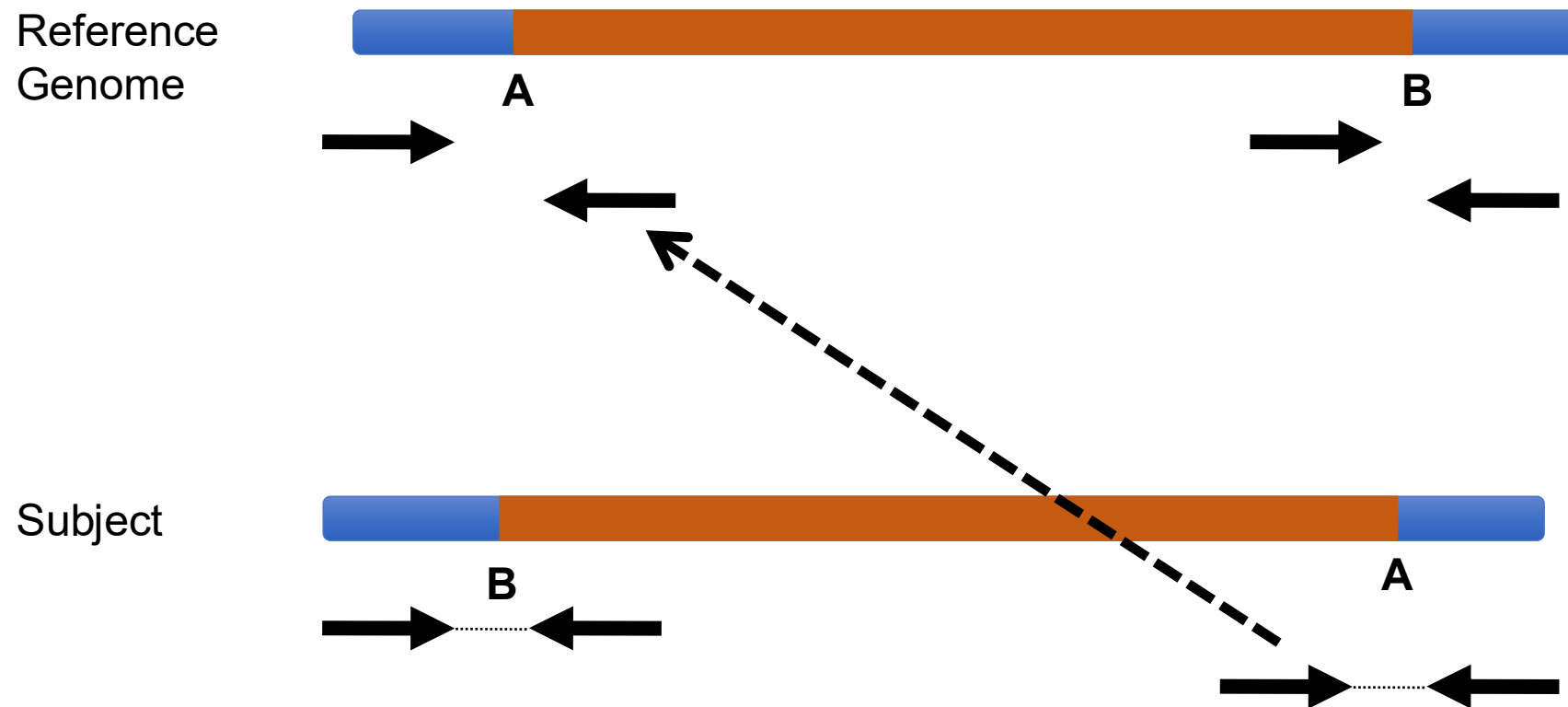
Subject



Inversion

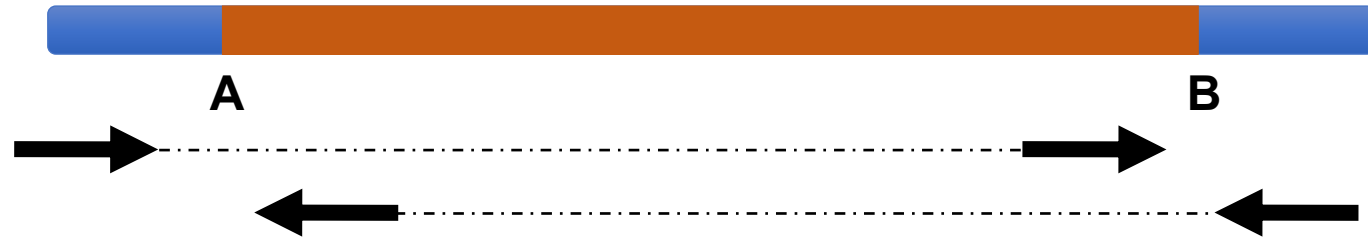


Inversion



Inversion

Reference
Genome



Inversion



Anomaly: expected orientation of pair is
inward facing ($\longrightarrow \longleftarrow$)

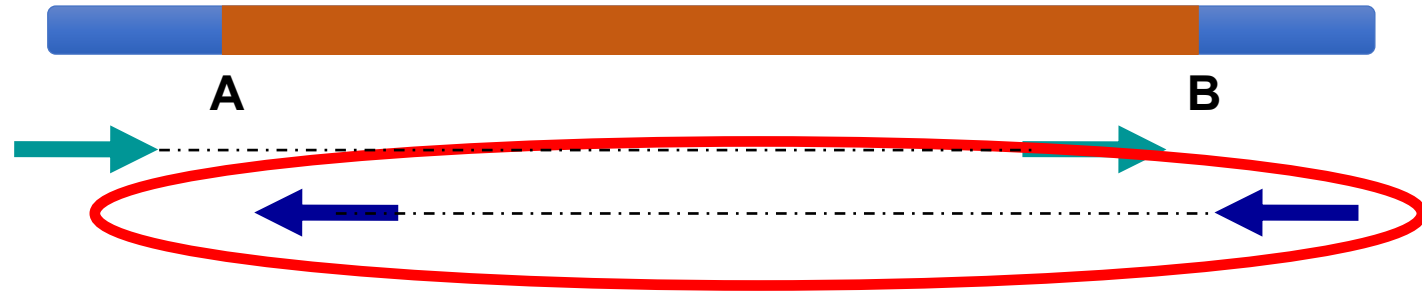
Inversion



“Left” side pair

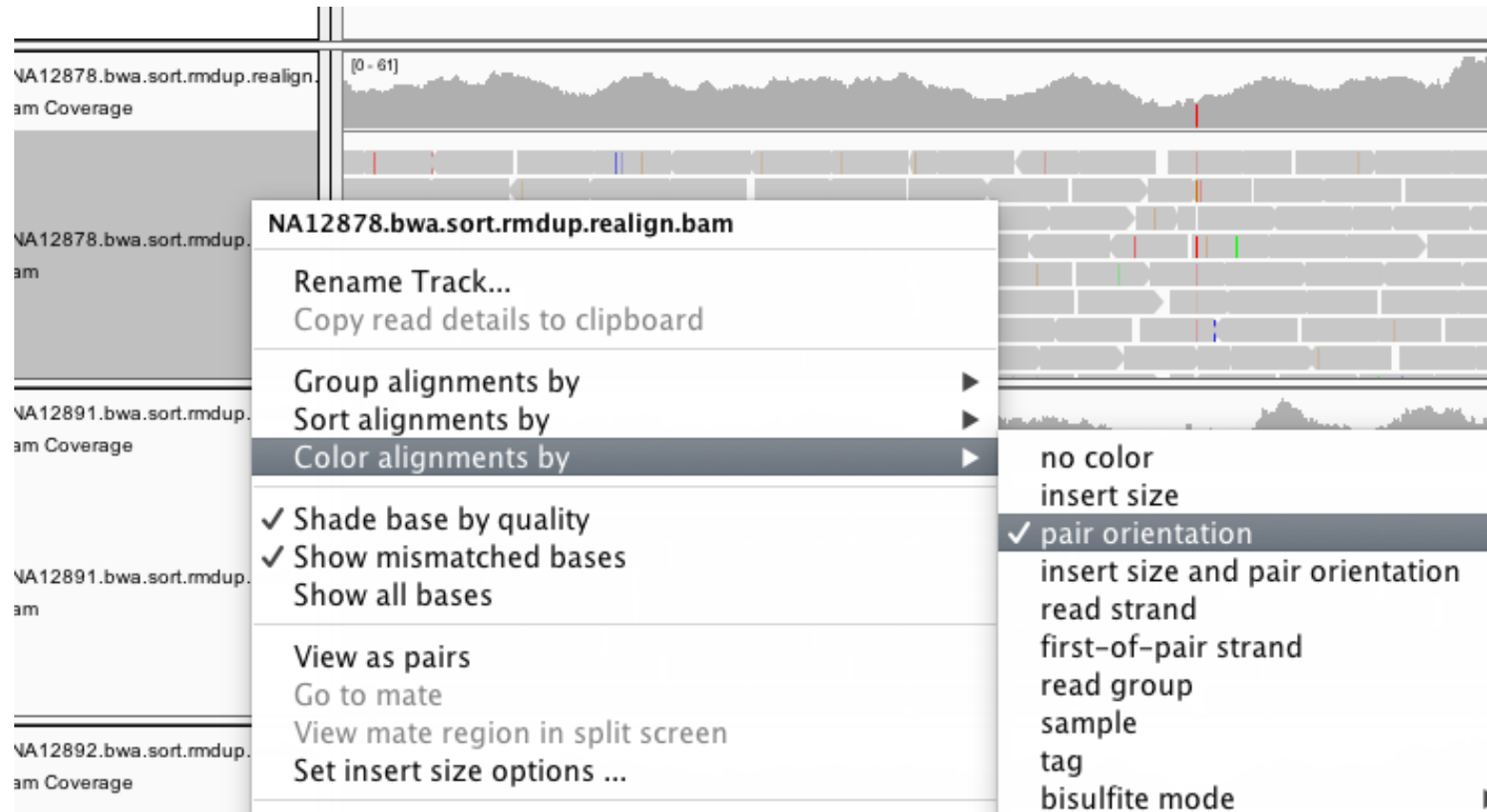
Inversion

Reference
Genome

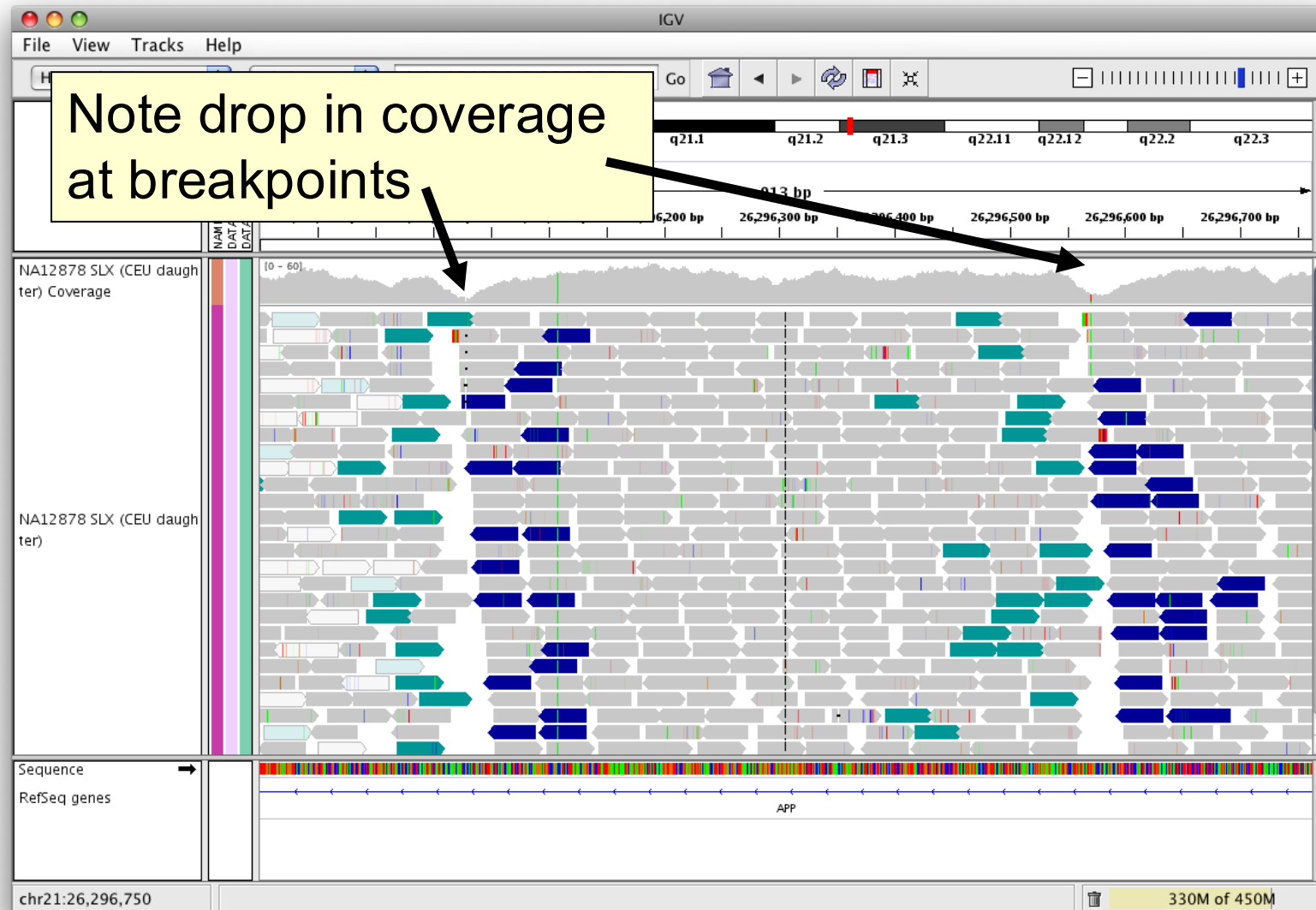


“Right” side pair

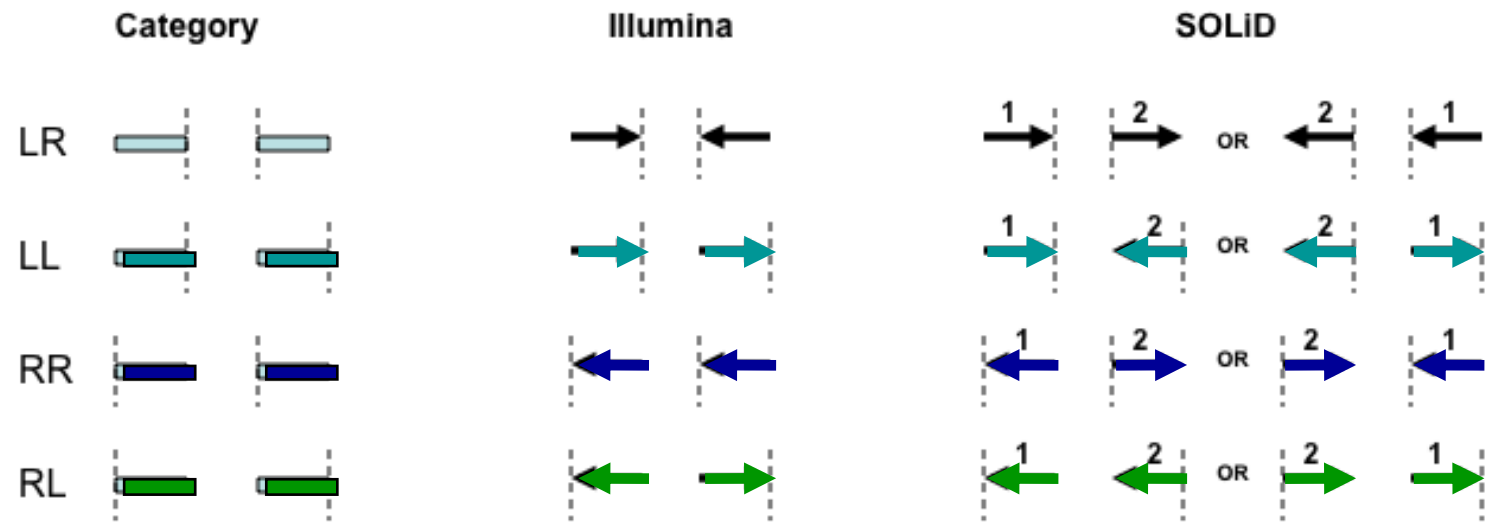
Color by pair orientation



Inversion



Interpretation of read pair orientations

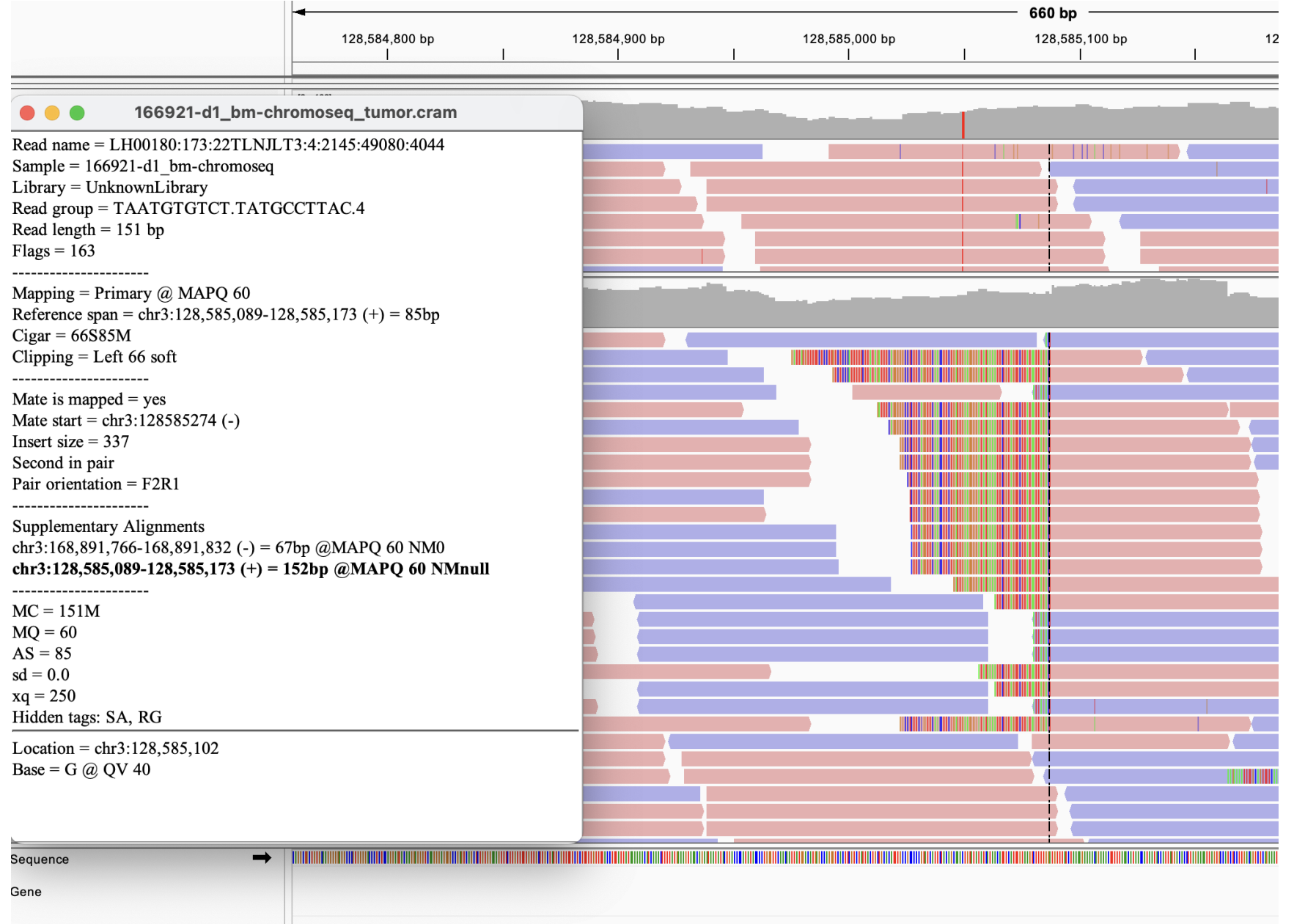


- LR Normal reads.
The reads are left and right (respectively) of the unsequenced part of the sequenced DNA fragment when aligned back to the reference genome.
- LL,RR Implies inversion in sequenced DNA with respect to reference.
- RL Implies duplication or translocation with respect to reference.

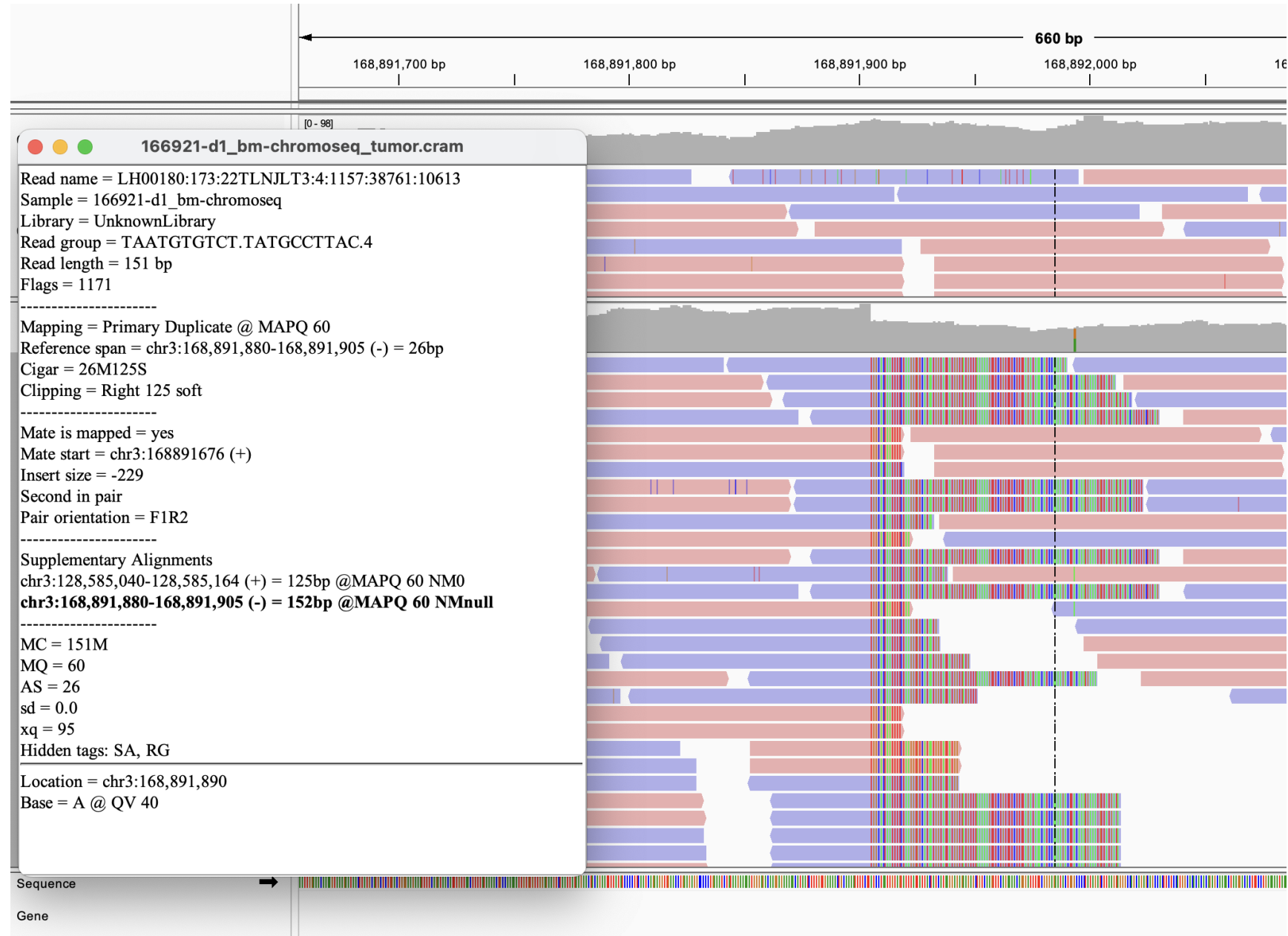
These categories only apply to reads where both mates map to the same chromosome.

Figure courtesy of Bob Handsaker

Softclipping and supplementary alignments



Softclipping and supplementary alignments



Softclipping and supplementary alignments

166921-d1_b

Read name = LH00180:173:22T
Sample = 166921-d1_bm-chron
Library = UnknownLibrary
Read group = TAATGTGTCT.
Read length = 151 bp
Flags = 1171

Mapping = Primary Duplicate @
Reference span = chr3:168,891,
Cigar = 26M125S
Clipping = Right 125 soft

Mate is mapped = yes
Mate start = chr3:168891676 (+
Insert size = -229
Second in pair
Pair orientation = F1R2

Supplementary Alignments
chr3:128,585,040-128,585,164
chr3:168,891,880-168,891,905

MC = 151M
MQ = 60
AS = 26
sd = 0.0
xq = 95
Hidden tags: SA, RG

Location = chr3:168,891,890
Base = A @ QV 40

Sequence
Gene

Re-pack alignments

☒ Shade base by quality
☒ Show mismatched bases
Show all bases
Duplicates
Hide small indels

View as pairs
Go to mate
View mate region in split screen
Set insert size options ...

Link supplementary alignments
Link by tag...
View chimeric alignments in split screen
Supplementary Reads Diagram

Collapsed
☒ Expanded
Squished
Full

Select by name...
Clear selections

Copy read details
Copy read sequence
Copy right-clipped sequence
Copy consensus sequence

BLAT read sequence
BLAT right-clipped sequence

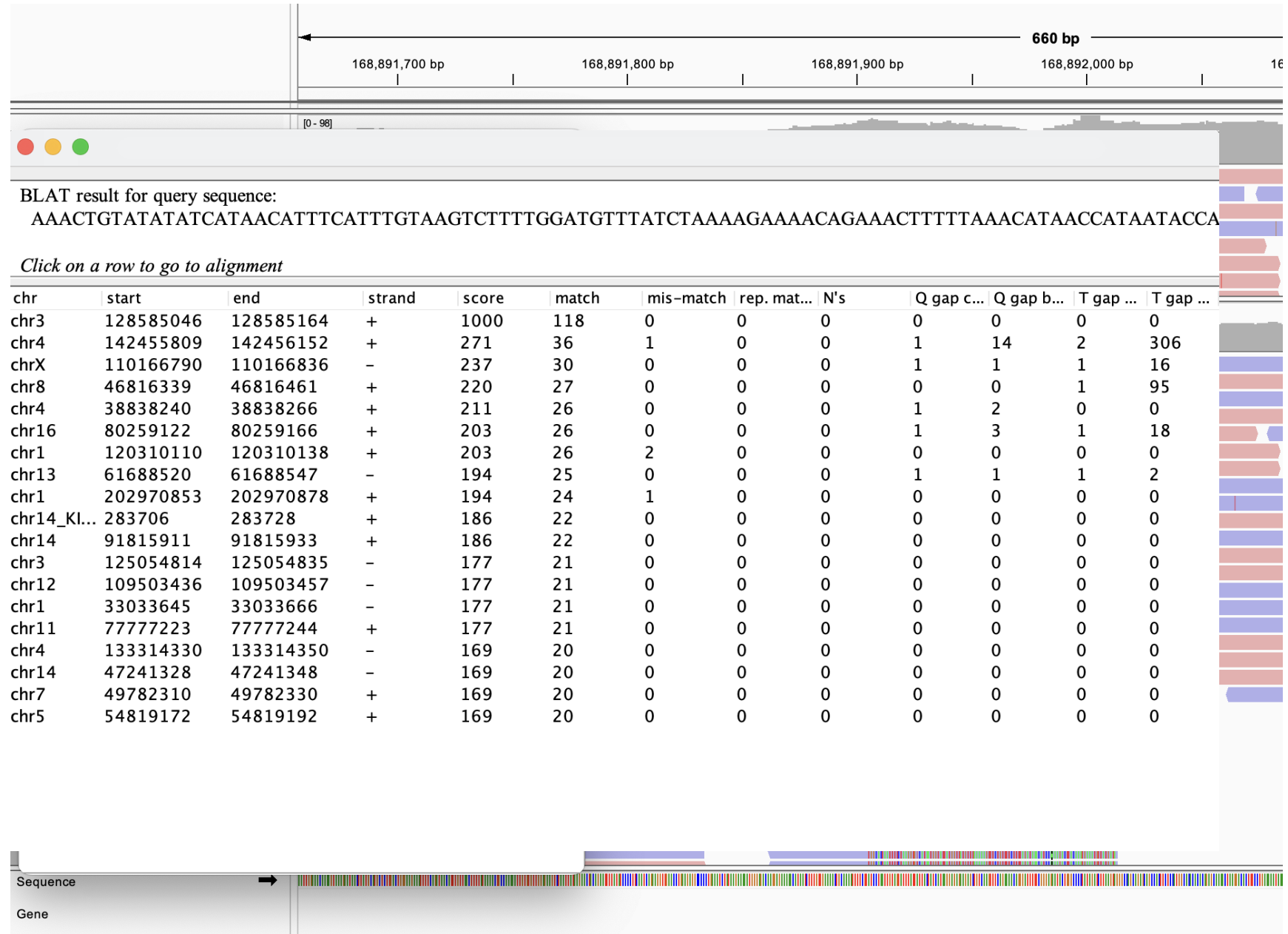
☒ Show Coverage Track
Show Splice Junction Track
☒ Show Alignment Track

Save PNG image...
Save SVG image

660 bp

168,891,900 bp168,892,000 bp16

Softclipping and supplementary alignments



Assignment

- <https://pmbio.org/module-03-align/0003/03/01/IntroToIGV/>